



**OSTIM TECHNICAL
UNIVERSITY
A N K A R A**

**T.C.
OSTIM TECHNICAL UNIVERSITY
Faculty of Engineering**

Department of Electrical and Electronics Engineering

**Autonomous Ground-Level Scanning Vehicle
Using Ultrasonic Radar and Laser Targeting
(Eye Of Horus)**

Graduation Project I and II

220202019 Ebru Karcı
210202996 Amr Ayman
210202943 Abdalla Hossni
210202885 Ahmed Mostafa
210202967 Ahmed Soliman

**Supervisor
Prof. Dr. İsmail Hakkı ALTAŞ**

May 2026
ANKARA



**OSTIM TECHNICAL
UNIVERSITY
ANKARA**

T.C.

**OSTIM TECHNICAL UNIVERSITY
Faculty of Engineering**

Department of Electrical and Electronics Engineering

**Autonomous Ground-Level Scanning Vehicle
Using Ultrasonic Radar and Laser Targeting
(Eye Of Horus)**

Graduation Project I and II

220202019 Ebru Karcı
210202996 Amr Ayman
210202943 Abdalla Hossni
210202885 Ahmed Mostafa
210202967 Ahmed Soliman

Supervisor

Prof. Dr. İsmail Hakkı ALTAŞ

May 2026
ANKARA

GRADUATION PROJECT APPROVAL FORM

The undergraduate graduation project titled “[Autonomous Ground-Level Scanning Vehicle Using Ultrasonic Radar and Laser Targeting]”, prepared by [Amr Ayman , Ahmed Mustafa , Ebru karcı , Ahmed Soliman , Abdalla hossni] under the supervision of [Prof. İsmail Hakkı ALTAŞ] was examined and accepted as an Undergraduate Graduation Project in terms of its scope and quality.

Supervisor : Prof. Dr. İsmail Hakkı ALTAŞ

Jury member 1 : Prof. Dr. V.Yuchenko

Jury member 2 : Dr. Hüseyin köse

Head of the Department : Prof. Dr. İsmail Hakkı ALTAŞ

FOREWORD

This Graduation Project has been prepared as part of the requirements of the Electrical and Electronics Engineering Department at OSTİM Technical University. The purpose of this work is to contribute to the development of autonomous sensing systems by designing and simulating a ground-level scanning vehicle that combines ultrasonic radar detection with laser-based visual marking.

We would like to express our deepest gratitude to our supervisor, Prof.Dr. İsmail Hakkı ALTAŞ, for his continuous guidance, technical insights, and encouragement throughout the development of this project. We also extend our thanks to the esteemed faculty members, laboratory technicians, and staff of the Engineering Faculty for their support and for providing the necessary resources during the design and simulation stages of this study.

We sincerely appreciate the cooperation, teamwork, and dedication of our project group members, whose combined efforts made this work possible. We also thank the Department Chair and the Engineering Faculty for enabling us to utilize departmental facilities during the preparation of this project. Additionally, we acknowledge the support of our families, who encouraged and motivated us during every stage of our academic journey.

This project represents not only a technical achievement but also a significant educational experience that strengthened our understanding of autonomous systems, embedded programming, and engineering design principles. We hope that this work contributes meaningfully to future studies and inspires other students working in similar fields.

May 2026

Ebru karcı

Amr Ayman

Abdalla Hossni`

Ahmed Mostafa

Ahmed Soliman

CONTENTS

Graduation Project Approval Form	i
Foreword	ii
Contents.....	iii
Abstract.....	iv
Symbols and Abbreviations	v
The list of figure	vi
The list of tables	vii
1. INTRODUCTION	1
1.1 General Information	1
1.2 Literature Review	1
1.3 Originality (Novelty)	2
1.4 Methodology	3
1.4.1. Idea Development and Literature Review.....	3
1.4.2. System Architecture Design.....	4
1.4.3. Detection Logic and Control Algorithm.....	4
1.4.4. Simulation Studies	4
1.4.5. Full Prototype.....	5
1.4.6. Documentation	5
1.5 Impact	5
1.6 Standards	6
1.7 Work Schedule	7
1.8 Risk Analysis	10
1.9 Organization of Work Packages	11
2. THEORETICAL BACKGROUND	13
2.1. General Information.....	13
2.2. Ultrasonic sensing theory	13
2.2.1. Working Principles.....	13
2.2.2. Time of flight measurement.....	13

2.2.3.	Advantage of ultrasonic sensor.....	13
2.2.4.	Limitation	14
2.3.	Laser Aiming module theory	14
2.3.1	Purpose of the leaser	14
2.3.2	Laser safety classes	14
2.3.3	Laser Alignment	14
2.4.	Arduino Microcontroller Theory	15
2.4.1	Why Arduino?	15
2.4.2.	Main Arduino task	15
2.4.3	Real-time Processing	15
2.4.4	Bluetooth Communication Theory	15
2.5.	Ground level detection theory	16
2.5.1	Horizontal plan detection	18
2.5.2	Detection zone.....	18
2.5.3	Control Response Mechanism.....	18
2.6.	System integration theory	19
2.6.1	Theoretical operation flow	20
2.6.2	Signal processing	20
2.6.3	Visual identification	20
2.7.	Summary	20
3.	DESIGN	21
3.1.	General Information.....	21
3.2.	Dimensioning.....	21
3.2.1.	Vehicle Chassis	22
3.2.2.	Sensor Mounting Height	23
3.2.3.	Servo-Sensor Mechanism.....	23
3.2.4.	Laser Alignment.....	24
3.2.5.	Laser Arm Design.....	24
3.2.6.	Laser Targeting Arm Mechanism	24
3.3.	System components and Their selections.....	25

3.4	Applied Methods.....	28
3.4.1.	Vehicle Movement	28
3.4.2.	Servo-Based Radar Scanning	28
3.4.3.	Object Detection Logic	28
3.4.4.	Continuous Loop	29
3.5.	Software used.....	29
3.6.	Component list and Economic analysis.....	30
3.7.	Legal Aspects.....	31
3.7.1.	Laser Safety (IEC 60825-1)	31
3.7.2.	Waste Electrical and Electronic Equipment (WEEE).....	31
3.7.3.	National Electronics Regulations	31
4.	SIMULATION STUDIES	32
4.1.	General information.....	32
4.2.	Simulation Environment	32
4.2.1.	Tinkercad Circuits.....	32
4.3.	Circuit design and wiring.....	33
4.4.	Simulation setup and initialization.....	34
4.5.	Simulation workflow.....	34
4.5.1.	Continuous Movement Mode.....	35
4.5.2.	Scanning Process	35
4.5.3.	Object Detection	35
4.5.4.	Manual Movement Mode	35
4.5.5	Real-Time Control Loop	35
5.	EXPERIMENTAL STUDIES	37
5.1.	General Information	37
5.2.	System Integration Overview.....	37
5.3.	Mechanical Assembly.....	37
5.4.	Electrical Connections.....	39
5.5.	Embedded System Implementation.....	40
5.6.	Prototype Operation.....	40

5.7. Results and Observations.....	41
6. RESULTS.....	42
6.1. Introduction.....	42
6.2. Simulation Results.....	43
6.3. Experimental Results.....	44
6.3.1. Ultrasonic Sensor Accuracy and Linearity.....	44
6.3.2. Operational Success Rate and Functional Validation.....	45
6.4. Comparison of Simulation and Experimental Results.....	46
6.5. Discussion.....	46
7. CONCLUSIONS.....	47
7.1 Future Work	47
8. REFERENCES.....	48
APPENDICES.....	49

ABSTRACT

This project presents the design, simulation, and preliminary validation of an *Autonomous Ground-Level Scanning Vehicle* that utilizes an ultrasonic radar mechanism combined with a laser targeting module to detect, identify, and visually mark obstacles within its path. The system integrates an Arduino-based microcontroller, a servo-mounted HC-SR04 ultrasonic sensor for angular scanning, a secondary servo for laser alignment, and supporting components including a buzzer, LCD display, and motor driver. A stop-and-scan operational strategy was implemented, in which the vehicle periodically interrupts its motion to perform a full angular sweep and evaluate the surrounding environment using time-of-flight (ToF) measurements.

In this study, the complete system architecture, control logic, and sensing behavior were designed and tested through Tinkercad simulations. The scanning mechanism, laser-based visual marking, and obstacle-locking behavior were successfully implemented, demonstrating coherent interaction among all subsystems. Simulation results confirmed accurate distance measurement, reliable servo positioning, effective laser-point indication, and stable decision-making under various detection scenarios.

The findings show that the proposed low-cost platform can effectively emulate radar-like detection principles using ultrasonic sensing, making it suitable for autonomous robotics education, indoor navigation research, and embedded systems training. The system also supports manual wireless control using an HC-05 Bluetooth module, allowing the user to override autonomous behavior and directly control the vehicle in real time. The simulation outcomes validate the feasibility of the design and prepare the foundation for full hardware implementation in chapter 5 experimental studies.

SYMBOLS AND ABBREVIATIONS

Symbol	Meaning
d	Measured distance (cm or m)
t	Time-of-flight (μs)
v	Speed of sound in air ($\approx 343 \text{ m/s}$)
θ	Servo rotation angle (degrees)
T	Trigger pulse duration (μs)
τ	Echo return time (μs)
DT	Detection threshold distance (cm)
As	Servo angle step (degrees)
Al	Locked detection angle (degrees)
IL	Laser indicator state (ON/OFF)
IB	Buzzer indicator state (ON/OFF)
V_{cc}	Supply voltage (+5V)
I	Electrical current (mA)
P	Power (W)
Abbreviation	Meaning
ToF	Time of Flight
HC-SR04	Ultrasonic distance sensor module
LCD	Liquid Crystal Display
LED	Light Emitting Diode
RF	Radio Frequency
PWM	Pulse Width Modulation
DC	Direct Current
I / O	Data in and data out.
CAD	Computer-Aided Design
GUI	Graphical User Interface
CNC	Computer Numerical Control
GND	Electrical Ground
VCC	Power Supply Voltage
UNO	Arduino Uno microcontroller board
MG90s	Micro Servo Motor
L298N	DC Motor Driver Module
IDE	Integrated Development Environment
Hz	Hertz (frequency unit)
kHz	Kilohertz
ms	Millisecond
μs	Microsecond
cm	Centimeter
HC-05	Bluetooth serial communication module

THE LIST OF FIGURE

- Figure 2.1** working of ultrasonic sensor
- Figure 2.2** Flowchart of the Autonomous Scanning Algorithm
- Figure 3.1** SolidWorks Assembly of the Autonomous Radar Vehicle
- Figure 3.2** Top View of Mechanical Layout
- Figure 3.3** Dimensions View of Mechanical Layout
- Figure 3.4** Dimensions View of laser arm design
- Figure 3.5** Arduino IDE interface showing the implemented control code
- Figure 4.1** Full Wiring Diagram of the Radar System
- Figure 4.2** Close-Up View of Ultrasonic Sensor Mounting in Simulation
- Figure 4.3** LCD Showing System Startup Message
- Figure 4.4** Real-Time Behavior of the Autonomous Ground
- Figure 5.1** Intermediate Stage of Prototype Development Showing Initial Assembly and Component Placement
- Figure 5.2** Bottom View of the Robotic Chassis Showing Power Source and Drive Train Integration
- Figure 5.3** Lateral View of the Integrated Prototype
- Figure 5.4** Final Integrated Autonomous Robotic System "Eye of Horus."
- Figure 6.1** Prototype During Operation Showing Obstacle Detection and Laser Targeting Toward the Detected Object
- Figure 6.2** Simulated Distance vs Measured Distance (Tinkercad)
- Figure 6.3** Experimental Results: Actual vs. Measured Distance
- Figure 6.4** System Functional Success Rate Chart.

THE LIST OF TABLES

Table 1.1	Summary of Literature Review Findings
Table 1.2	Most Important TSE, IEC, and IEEE Standards
Table 1.3	Work-Time Table
Table 1.4	Risk analysis and plan B
Table 1.5	Work packages
Table 3.1	Economic Cost Analysis of Components
Table 4.1	Simulation Scenarios and Outputs
Table 4.2	LCD Display States During Simulation
Table 4.3	Object Detection and Laser Alignment Data
Table 6.1	Simulated Distance Measurements in (Tinkercad)
Table 6.2	Experimental Sensor Accuracy Data
Table 6.3	System Functional Success Rate (15 Trials)

1. INTRODUCTION

1.1 General Information

This project presents the design and implementation of an autonomous ground-level scanning vehicle that detects obstacles using an ultrasonic radar mechanism and visually marks their positions using a laser aiming module. **The system integrates an Arduino microcontroller, DC motors, an ultrasonic sensor mounted on a servo, a laser-pointer module, and supporting components such as a buzzer and LCD display.** Unlike traditional radar systems that operate with electromagnetic waves, **this vehicle utilizes ultrasonic time-of-flight measurements to construct a simplified radar-like detection system suitable for short-range sensing.** The robot operates in both autonomous and manual modes. In autonomous mode, the vehicle performs obstacle detection and targeting automatically, while in manual mode the user can control the movement wirelessly through Bluetooth communication. and performs periodic stop-and-scan cycles. The vehicle temporarily hold, stabilizes its sensors, and performs a full angular sweep to detect objects along the ground plane. **If an object is detected, the vehicle stops, the buzzer alerts, and the arm of laser move toward the detected target's position on the ground.**

This approach provides a low-cost, educational demonstration of autonomous navigation, embedded sensing, and radar-inspired detection principles. The system is designed to be simple, modular, and suitable for laboratory use, robotics instruction, and sensor-technology training.

1.2 Literature Search

Ground-level object detection is widely discussed in robotics research, autonomous vehicles, short-range sensing platforms, and intelligent monitoring systems. Many studies emphasize the importance of low-cost sensors for navigation and obstacle detection in mobile robots, particularly for indoor environments and educational applications [4], [5], [6].

Ultrasonic sensors have gained significant attention due to their affordability, robustness in short-range environments, and insensitivity to lighting conditions. Their operating principle is based on transmitting a high-frequency (≈ 40 kHz) sound pulse and calculating distance from the reflected echo. Numerous studies confirm their suitability

for ground-level mobile robotics due to their high reliability and ease of integration with microcontrollers [4], [5], [6].

Arduino microcontrollers are commonly used in embedded robotics research because of their flexible I/O structure, open-source software ecosystem, and ease of interfacing with servos, motors, and sensors. Prior research highlights Arduino's strength in implementing real-time decision-making, sensor fusion, and data acquisition for autonomous robot systems [8], [9].

Laser modules have also been used in various academic studies as visual indicators, target-marking mechanisms, and alignment tools. When combined with ultrasonic detection, laser pointing provides intuitive user feedback by highlighting the detected position of an object in real time [7], [10].

Although classical radar systems rely on radio-frequency waves, several academic works show that ultrasonic waves can replicate essential radar concepts—signal transmission, reflection, and time-of-flight measurement—at short distances and significantly lower cost. These “ultrasonic radar analogues” are especially common in robotics training environments [4], [5], [8].

Overall, the literature supports the following elements of this project:

1. **Ultrasonic sensors** for short-range obstacle detection [4], [5], [6].
2. **Arduino microcontrollers** for real-time processing and autonomous control [8]
3. **Laser modules** for intuitive visual indication of detected targets [7], [10].
4. **Radar-inspired time-of-flight methods** adapted for low-cost mobile robotics [4], [5], [8].

This project synthesizes these components into a unified autonomous ground-level scanning vehicle capable of real-time target marking.

1.3 Originality (Novelty)

The originality of this project lies in integrating ultrasonic radar scanning with laser-based ground-level visual indication within a fully autonomous mobile vehicle. While ultrasonic sensors and laser modules have been used independently in several studies, their combined use in a stop-and-scan autonomous platform is not commonly found in academic literature. Unlike many robotic vehicles that only detect obstacles directly in front, the proposed system performs a periodic angular sweep

using a servo-mounted ultrasonic sensor. This scanning method **provides a radar-like 2D detection** profile of the surrounding ground environment. When an object is detected, the secondary servo aligns the laser pointer precisely to the detected angle, allowing visual marking of the target's location.

The project is also novel in its educational focus, offering a clear and intuitive visualization of detection events. The combination of mobile motion, autonomous decision-making, radar-style scanning, and laser marking creates an instructional system that helps students understand embedded sensing, robotics, and time-of-flight measurement concepts. The system uses only low-cost components, making it more accessible compared to traditional radar platforms.

Although individual hardware components are well-known, the integrated structure, unique scanning behavior, and visual marking technique provide originality appropriate for a senior graduation project. The integration of autonomous scanning with real-time manual wireless override increases the flexibility and interactivity of the system.

1.4 Methodology

The methodology of this study follows a structured engineering workflow including idea development, literature analysis, system design, simulation, and prototype implementation. Each step was planned based on the Graduation Project Template to ensure technical adequacy and reproducibility.

1.4.1. Idea Development and Literature Review

The project idea was developed after reviewing academic studies related to ultrasonic sensing, radar principles, mobile robotics, and laser-based visualization techniques. Previous research on ultrasonic-based obstacle detection and autonomous robotic navigation provided the theoretical foundation for the proposed system [4], [5], [6], [8]. These studies demonstrate that ultrasonic sensors combined with microcontroller-based control systems can effectively perform obstacle detection and navigation tasks in robotic platforms. The main findings from the reviewed literature and their relevance to the proposed system are summarized in Table 1.1.

TABLE 1.1 Summary of Literature Review Findings

Topic	Key Findings	Relevance to Project	Reference
Ultrasonic Sensing	Reliable for short-range ground-level detection; unaffected by lighting; low cost.	Used as the main detection sensor (HC-SR04).	[4], [6]
Radar Principles	TOF-based detection is similar between RF radar and ultrasonic sonar.	Basis for “radar-like” behavior in the system.	[4], [5]
Arduino Systems	Widely used for real-time sensing and control.	Arduino Uno processes scanning, detection, and laser activation.	[8], [9]
Laser-Based Indicators	Simple, accurate visual target indicators.	Laser used to point at the detected object angle.	[7], [10]
Robotics Navigation	Stop–scan–decide movement improves reliability.	System uses autonomous movement cycle with scanning intervals.	[5], [8]

1.4.2. System Architecture Design

The hardware architecture includes:

- An **ultrasonic sensor** mounted on a rotational servo for scanning,
- An **Arduino microcontroller** for processing and decision-making,
- **DC motors with an L298N driver** for autonomous motion,
- A **laser module** mounted on a second servo for target marking,
- Supporting components such as an LCD and buzzer.
- HC-05 Bluetooth module for wireless manual control

1.4.3. Detection Logic and Control Algorithm

The vehicle drives forward autonomously. Every 0.5 seconds, it stops and performs a full radar sweep (0° – 180°). The Arduino calculates distance using time-of-flight and checks whether an object lies within a specified threshold. If detected, the system stops movement, aims the laser pointer toward the object angle, activates the buzzer, and displays information on the LCD.

1.4.4. Simulation Studies

Simulations using **Tinkercad** were conducted to verify sensor timing, servo movement, and obstacle detection logic. Mechanical structure and component arrangement were designed and validated using SolidWorks models.

1.4.5. Full Prototype

The complete mobile prototype will be assembled. Tests will evaluate detection accuracy, scanning reliability, laser pointing precision, and motion stability under different surface and lighting conditions.

1.4.6. Documentation

All stages conceptual design, simulation, system architecture, detection logic, experiments, and results are documented per the project guidelines to ensure clarity and reproducibility.

This methodology creates a clear pathway from initial concept to final autonomous vehicle implementation.

1.5 Impact

This project contributes to the fields of robotics education, autonomous systems, and ground-level sensing. By combining ultrasonic scanning with laser-based visual marking, the system provides an intuitive and demonstrative way to understand object detection concepts. The autonomous movement and periodic scanning approach illustrate real-world behaviors used in smart robots, warehouse vehicles, and automated security platforms. The manual control feature also improves training, testing, and educational usability.

The design can be improved and expanded into:

- Mobile robotic navigation systems,
- Autonomous ground vehicles (AGVs),
- Industrial perimeter monitoring,
- Indoor mapping and scanning robots,
- Teaching tools for sensing principles and embedded programming.

Because the system is built with low-cost, widely available components, it provides an accessible platform for academic training, laboratory demonstrations, and future student research projects in autonomous robotics.

1.6 Standards

To ensure the safety, reliability, and proper functioning of the stationary ground-level radar system, it is essential to follow recognized international and national standards. Because the project incorporates laser modules, ultrasonic sensors, and Arduino-based electronic control, compliance with key **IEC**, **TSE**, and **IEEE** standards is required. These standards regulate laser safety, electromagnetic compatibility, sensor calibration, and microcontroller–sensor communication. In table 1.2 summarizes the most critical standards relevant to the design and implementation of this system.

Table 1.2 Most Important TSE, IEC, and IEEE Standards

Standard Body	Standard Code	Standard Name	Relevance to the Project
IEC	IEC 60825-1	Safety of Laser Products	Ensures safe use and classification of the laser aiming module.
TSE (TS EN)	TS EN 60825-1	Laser Safety – Turkish Standard	Local compliance requirement for laser-based devices.
IEC	IEC 61131-2	Programmable Controllers – Requirements	Applies to Arduino’s function as a control unit.
IEC	IEC 61000-6-2 / 61000-6-4	EMC Immunity & Emission Standards	Ensures the system resists interference and does not cause electromagnetic disturbance.
ISO/IEC	ISO/IEC 16063-11	Calibration of Ultrasonic Sensors	Ensures accurate distance measurement and sensor performance.
IEC	IEC 61010-1	Safety for Measurement & Control Equipment	Ensures safe operation of low-voltage electronic circuits.
IEEE	IEEE 1451	Smart Transducer Interface Standards	Defines communication and interface requirements between sensors and the microcontroller.

1.7 Work Schedule

A work plan is placed at the end of the introduction under this heading. This work plan is organized as a work-time chart. It briefly summarizes who will take part in the work packages defined in the work-time graph and what will be done. What will be achieved when each work package is completed is briefly explained in a few sentences. The tasks in the work packages are shared so that each student in the team is the leader in at least one work package. The student responsible for a work package monitors the work of the other students and ensures that the work package is completed in the planned time with the planned result. In case there is a disruption in the processes to be carried out in the work packages, a plan B should be created and a plan B should be attached to each work package to ensure that progress is not interrupted. A B plan is not required for work packages where it is certain that there will be no disruption. However, a plan B should be created for situations where disruption may occur and the project should be completed within the specified time. A sample Work-Time chart is given in Table 1.3.

Table 1.3. Work-Time Table

WP No	Name and Purpose of the Work Package	Working Team	Time Interval	Contribution to the Project
1	Creating the project idea, conducting literature research, and defining the system concept.	Ahmed Mostafa (L) Amr Ayman Abdullah Hosny Ahmed Soliman Ebru Karci	October 2025	The project idea was created. Literature scanning is completed. Fundamental understanding of ground-level detection systems was formed. The purpose of using ultrasonic scanning and laser indicated zone detection was clarified
2	Determination of the working principles, selection of methods, and transition to the design.	Amr Ayman(L) Ahmed Mostafa Abdullah Hosny Ahmed Soliman Ebru Karci	October- November 2025	The scanning mechanism, target-detection system, and communication structure were determined. These decisions form the theoretical basis for design.
3	Completion of design calculations, component selections, budgeting, and preparation of system schematics	Ebru Karci (L) Amr Ayman Abdullah Hosny Ahmed Soliman Ahmed Mostafa	November- December 2025	Required Component choices (microcontroller, sensors, motors, communication module, power system) were selected, required Arduino codes were developed, and the system schematic was drawn.
4	Creating the simulation model, performing simulations, evaluating the results and writing the Graduation Project book	Abdullah Hosny (L) Amr Ayman Ahmed Mostafa Ahmed Soliman Ebru Karci	December 2025	Basic system behavior is tested in simulation (zone detection routine + laser activation logic). Early software structure is implemented and validated.

5	Final exams and Graduation Project presentations,	Ahmed Soliman (L) Amr Ayman Abdullah Hosny Ahmed Mostafa Ebru Karci	December- January 2026	The Engineering Design Report and presentation were completed according to project standards. Evaluated by the jury during the presentation. This stage finalizes Graduation Project I requirements.
6	production and starting prototype production	Ebru Karci (L) Amr Ayman Abdullah Hosny Ahmed Soliman Ahmed Mostafa	February 2026	Electronic, mechanical and structural components required for the mobile robot platform and scanning unit were purchased. Material procurement enables the beginning of prototype construction
7	Creation of electrical electronic circuits required for prototype manufacturing in accordance with the design	Abdullah Hosny (L) Ahmed Mostafa Amr Ayman Ahmed Soliman Ebru Karci	February- March 2026	Arduino-based control system, ultrasonic sensor interface, motor driver connections, and power distribution circuits were assembled in accordance with the design. This stage is critical for enabling the next steps.
8	Completion of prototype assemblies and tests, evaluation of test results	Ahmed Mostafa (L) Amr Ayman Abdullah Hosny Ahmed Soliman Ebru Karci	April-May 2026	The robot platform, ultrasonic scanning part, laser module and communication system were fully assembled and tested. The performance, scanning stability and control response were evaluated.
9	Writing and submission of the Graduation Project book.	Ahmed Soliman (L) Amr Ayman Abdullah Hosny Ahmed Mostafa Ebru Karci	May 2026	All design; simulation, implementation and test results were documented according to the graduation project format.
10	Final exams, Graduation Project Exhibition and presentations and completion of the project	Amr Ayman (L) Ahmed Mostafa Abdullah Hosny Ahmed Soliman Ebru Karci	May 2026	The prototype made in the Graduation Project Exhibition is exhibited in working condition and evaluated by the Jury. A passing score is required.

1.8. Risk Analysis

Table 1.4. Risk analysis and plan B

WP No	Contribution to the Project	Risk Analysis
1	The project idea is defined, and the general scope and requirements are clarified through literature research. This stage guides all later design and development steps.	If the literature review is incomplete or the topic is too broad, the project may lack focus. As a backup, the scope can be narrowed and additional sources such as previous theses, IEEE papers, and open-source
2	The working principle of the autonomous ground-level scanning vehicle is determined. radar-style ultrasonic detection are defined.	If the selected detection strategy becomes too complex or inefficient, it can be simplified by reducing scanning angles or increasing delay
3	Hardware components (ultrasonic sensor, servos, laser module, motor driver, CD motors, LCD, buzzer, and Arduino) are selected and their initial placement on the	If a component is unavailable or incompatible during integration, an equivalent part with similar specifications will be substituted
4	A preliminary system model is created, and simulations are performed in Tinkercad to test ultrasonic readings, servo motion, and detection logic before prototype construction.	If the simulation does not accurately reflect real conditions or access to simulation software becomes limited, basic bench-testing with hardware components will be performed as an
5	Mechanical design and system layout are created in SolidWorks to visualize the final autonomous scanning vehicle structure.	If the mechanical model requires changes or conflicts during actual assembly, the design will be modified and simplified without affecting
6	The Engineering Design report is prepared and evaluated by the academic jury as part of Graduation Project I requirements.	Failure to meet formatting or reporting rules may result in rejection. Following the Graduation Project Writing Guide strictly ensures compliance and prevents academic
7	Required materials are purchased and preparations for prototype production begin.	If certain materials cannot be sourced, they will be ordered online from trusted vendors or replaced with
8	Electrical wiring and circuit connections of all modules (motors, sensors, servos, laser, buzzer, LCD) are completed according to the design.	Incorrect wiring may cause system malfunction. Support may be requested from laboratory technicians or instructors in case of difficulties.
9	Prototype assembly is completed, and functional tests are performed to verify autonomous movement, scanning accuracy, and laser pointing performance.	If the prototype does not perform as expected, scanning speed, threshold distance, servo angles, and motor speeds will be tuned to reach

1.9. Organization of Work packages

The project is structured into ten Work Packages (WPs). The first five work packages focus on **Engineering Design**, while the remaining five are dedicated to the **Project Implementation and Testing** phases. To ensure effective teamwork and balanced responsibility distribution, each student assumes the role of **Work Package Leader** for at least one WP. The leader of each WP is responsible for coordinating workflow, assigning tasks to the other group members, monitoring progress, and ensuring the timely completion of all deliverables.

Leadership rotation ensures equal participation and accountability among all team members. Each leader communicates expectations, sets internal deadlines, and provides continuous feedback. Under this sub-section, details about **who is responsible for each work package, their assigned tasks, and estimated completion times** are provided.

Team Management Principles

- Every student acts as a leader for at least one Work Package (WP), ensuring equal participation, responsibility, and development of leadership skills among all team members.
- The WP leader is responsible for defining task objectives, assigning responsibilities to supporting members, monitoring progress, and ensuring effective communication within the team and alignment with overall project goals.
- Supporting members carry out the subtasks assigned by the WP leader, actively contributing to design, implementation, testing, and documentation processes while maintaining coordination with the team.
- Weekly internal review meetings are conducted to evaluate progress, identify challenges, and ensure that all tasks are completed according to the planned schedule and project milestones.
- Collaborative tools, such as shared folders, cloud storage systems, and group communication platforms, are used to facilitate continuous coordination, efficient file sharing, and real-time communication among team members.

2. THEORETICAL BACKGROUND

2.1. General Information

Radar (**Radio Detection and Ranging**) is a detection system that determines the distance, position, or movement of an object by transmitting a wave and measuring the time required for the reflected signal to return. Classical radar systems operate using electromagnetic radio waves; however, the core detection principle can be implemented with other wave types. Although the system developed in this project does not utilize radio-frequency waves, it mimics the essential radar mechanism using ultrasonic sound waves. The basic radar concept consists of two key elements:

Inverse-Square Propagation

A transmitted wave travels outward, interacts with an obstacle, and returns to the sensing unit as an echo. The received signal provides information about the object's distance.

Time of Flight (ToF)

The distance to the object is calculated by measuring:

$$\text{Distance} = \frac{v \cdot t}{2} \quad (2.1)$$

Where:

v = speed of the wave (340 m/s for sound in air)

t = round-trip time of the wave

Equation (2.1) represents the time-of-flight principle used to calculate distance. The ultrasonic wave travels to the object and returns to the sensor, and the total travel time is measured. Since this time corresponds to a round-trip path, it is divided by two to determine the actual distance.

2.2 Ultrasonic Sensing Theory

Ultrasonic sensors measure distance using the time-of-flight principle, where a high-frequency sound wave is transmitted toward an object and reflected back to the receiver. The time taken for the echo to return is used to calculate the distance. This operating mechanism is illustrated in Figure 2.1.

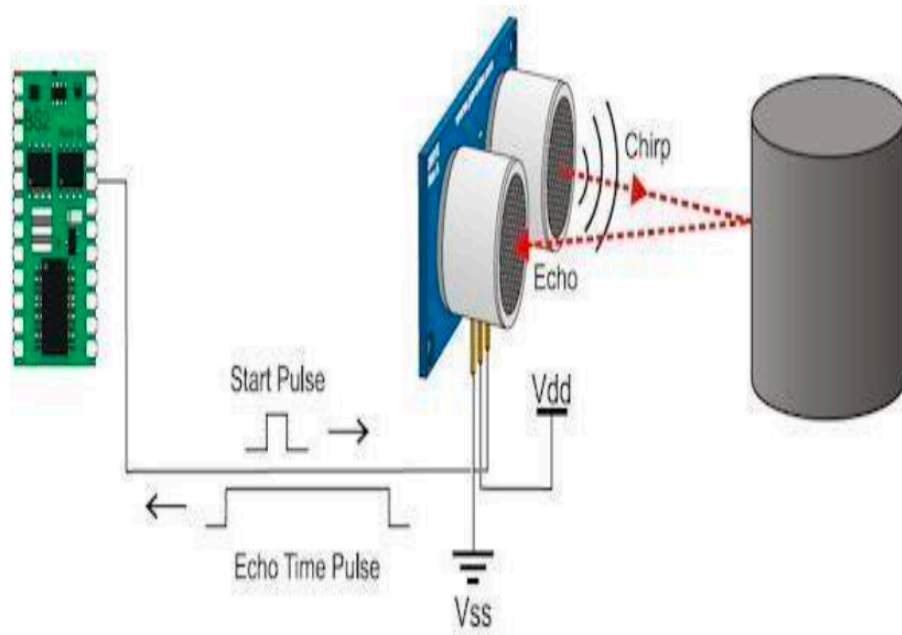


Figure 2.1. working of ultrasonic sensor

2.2.1 Working Principles

The HC-SR04 ultrasonic sensor consists of:

- **Transmitter:** emits 40 kHz pulse
- **Receiver:** listens for the returning echo
- **Control electronics:** measures echo duration

2.2.2 Time of Flight Measurement

The sensor outputs a pulse whose width corresponds to the distance. Arduino measures the pulse width using the pulseIn() function.

2.2.3 Advantages of Ultrasonic Sensors

- Low-cost
- Easy to interface
- Accurate for short-distance ground detection

2.2.4 Limitations

- Reflective errors from uneven ground
- Limited maximum range (~4 m)
- Soft materials absorb sound
- Environmental noise affects readings

These limitations explain why the system is used for **ground-based, short-range detection**.

2.3 Laser Aiming Module Theory

A laser module emits a narrow, focused beam of light. In this project, the laser serves as a **visual indicator** to show the position of the detected object on the ground.

2.3.1 Purpose of the Laser

- Enhance user visualization
- Show the exact target location
- Provide instant feedback

2.3.2 Laser Safety Classes

Most Arduino laser modules are **Class 2 or Class 3A**, requiring:

- No direct eye exposure
- Use of proper handling and caution

2.3.3 Laser Alignment

The laser must be aligned parallel to the ultrasonic sensor so both measure in the same

2.4 Arduino Microcontroller Theory

The Arduino microcontroller serves as the central processing unit and primary control element of the autonomous ground-level scanning vehicle. It functions as the **“brain” of the system**, coordinating all sensor readings, decision-making processes, and actuator responses. At its core, Arduino is a microcontroller development platform built on the ATmega family, offering digital I/O pins, analog input capability, pulse-width modulation outputs, timers, and serial communication interfaces. These features allow it to reliably interface with a variety of sensors and actuators used in autonomous robotic applications.

2.4.1 Why Arduino?

- Simple programming (C++ style)
- Fast enough for ultrasonic ToF measurements
- Low power consumption
- Plenty of GPIO pins
- Good for academic prototype systems

2.4.2 Main Arduino Tasks

1. Trigger ultrasonic pulses
2. Measure echo duration
3. Convert time to distance
4. Activate laser when object is detected
5. Display or log results

2.4.3 Real-time Processing

The Arduino processes ultrasonic readings continuously (loop function), enabling real-time object detection.

2.4.4 Bluetooth Communication Theory

- HC-05 overview
- Serial communication
- Wireless command transmission
- Manual control logic

2.5 Ground-Level Detection Theory

In this project, the radar system operates using a servo-mounted ultrasonic sensor that performs angular scanning across a horizontal plane. Unlike fixed forward-facing detection systems, the sensor rotates between defined angles (0° – 180°), allowing the system to monitor a wider ground-level area. This scanning approach enables the detection of objects at different directions relative to the vehicle.

The detection process is based on interpreting distance measurements obtained from the ultrasonic sensor at each scanning angle. The system evaluates whether an object enters a predefined threshold zone. When an object is detected, the microcontroller triggers corresponding control actions such as stopping the vehicle, aligning the laser toward the detected angle, and activating warning indicators.

2.5.1 Horizontal Plane Detection

Horizontal plane detection refers to the ability of the ultrasonic sensor to identify objects that are located within the same horizontal level as the sensor itself. Because ultrasonic sensors emit sound waves in a cone-shaped pattern, they create a measurable detection plane parallel to the ground. Any object intersecting this plane will reflect the sound waves back to the sensor, allowing the system to calculate the distance based on time-of-flight.

In this system, the ultrasonic sensor is mounted at a fixed height on the vehicle, meaning it is optimized to detect ground-level obstacles such as walls, boxes, stationary objects, or elevated edges. The sensor continuously emits pulses along the forward horizontal direction, forming a stable detection zone directly ahead of the vehicle. This ensures:

- **Consistent object detection at predictable heights**
Objects must lie approximately within the height of the sensor for accurate reflection.
- **Reliable detection of obstacles in front of the vehicle**
The horizontal plane aligns with the typical placement of physical obstructions, improving detection accuracy.
- **Reduced interference from ground reflections**
Because the sensor is slightly angled or elevated above the ground, unwanted reflections from the surface are minimized.
- **Continuous scanning without mechanical rotation**
The fixed horizontal plane simplifies operation while still providing effective forward monitoring.

2.5.2 Detection Zone

The operational logic of the autonomous scanning system follows a structured decision-making sequence that includes initialization, scanning, object detection, and response actions. This process flow is illustrated in Figure 2.2.

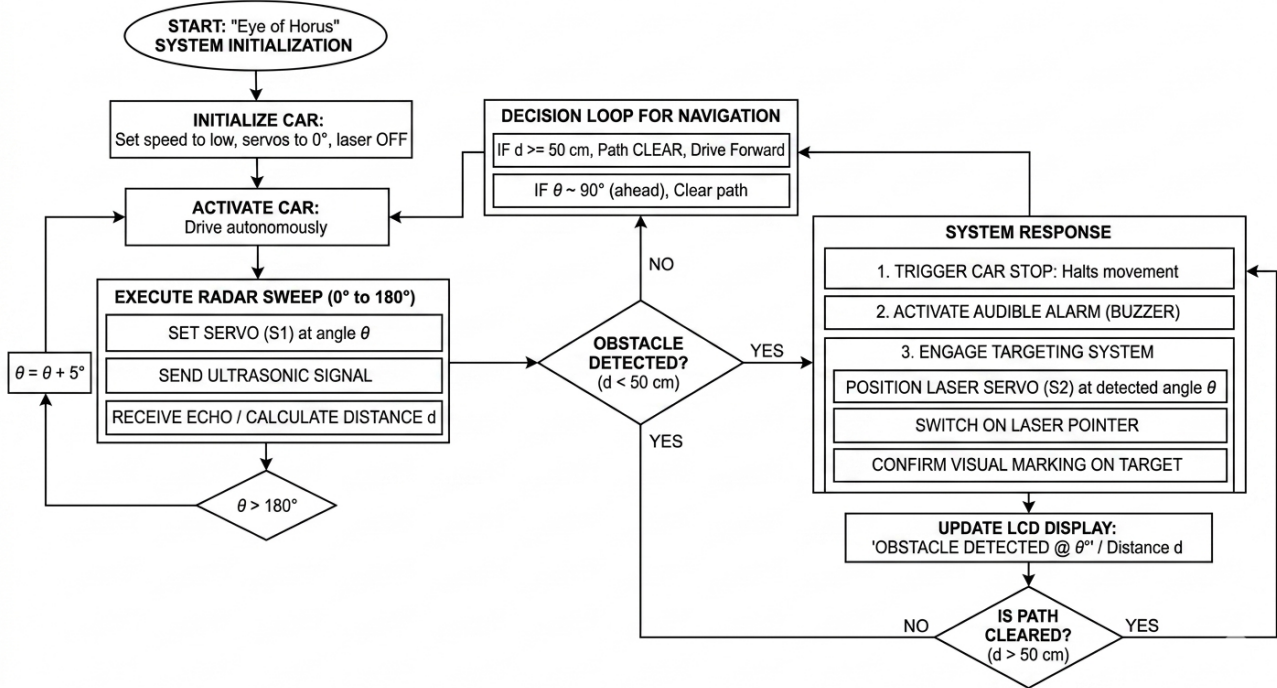


Figure 2.2. Flowchart of the autonomous scanning algorithms

2.5.3 Control Response Mechanism

When an object is detected within the unsafe zone, the microcontroller immediately triggers a control response sequence:

- Vehicle motion is interrupted by stopping the motors.
- The system may activate warning signals such as a buzzer or LCD alert.
- Servo-based components (if integrated) may align toward the detected direction for inspection.
- The system remains in a safe state until the obstacle is removed or the path is cleared.

This response ensures collision avoidance and provides real-time safety control for autonomous operation.

2.6 System Integration Theory

System integration in this project refers to the coordination of the mechanical, electronic, sensing, and wireless communication subsystems into a unified operational cycle. The system operates using both autonomous and manual control modes to achieve flexible navigation, environmental monitoring, and real-time obstacle detection.

During autonomous operation, the vehicle moves forward while continuously monitoring the surrounding environment using the ultrasonic and IR sensors. The ultrasonic sensor performs continuous distance measurement and scanning, while the IR sensors provide short-range collision detection to improve navigation safety. The LCD module displays the current operational status such as “MOVING,” “SCANNING,” or “OBJECT DETECTED.”

When an object is detected within the predefined threshold distance, the motors are temporarily stopped to stabilize the system and improve measurement accuracy. The servo-mounted ultrasonic sensor then performs angular scanning between 0° and 180° to determine the direction and distance of the detected object. The measured data is processed by the Arduino microcontroller and displayed on the LCD screen.

After the object position is confirmed, the system enters the targeting phase. In this phase, the robotic arm aligns the laser module toward the detected angle using servo-controlled movement. The laser pointer is activated to visually indicate the target position, while the buzzer generates an audible warning signal. The vehicle remains stationary during this phase to maintain stable laser alignment and accurate targeting performance.

The system also supports manual wireless control using the HC-05 Bluetooth module. Through a smartphone application, the user can control vehicle movement in real time using forward, backward, left, and right commands. Manual commands have higher priority than the autonomous detection logic, allowing the operator to override the system immediately whenever required.

After the object is removed or the targeting process is completed, the laser and buzzer are deactivated, the robotic arm returns to its default position, and the vehicle resumes environmental monitoring and movement. This integrated operational cycle demonstrates coordinated interaction between sensing, processing, wireless communication, and actuation systems within a compact autonomous robotic platform.

2.6.1 Theoretical Operation Flow

- Arduino commands ultrasonic sensor to transmit a sound pulse
- The pulse travels outward and reflects off objects
- The echo returns to the receiver
- Arduino computes distance using ToF
- During scanning, servo sweeps horizontally
- If an object is detected within a threshold, the laser aims toward the detected angle

2.6.2 Signal Processing

Ultrasonic sensor readings are subject to noise and fluctuations due to environmental factors such as surface texture, angle of incidence, and air disturbances. To improve measurement reliability, basic signal processing techniques are applied:

- **Averaging:** Multiple readings are taken and averaged to reduce random noise.
- **Threshold Filtering:** Only values within a valid range are considered.
- **Outlier Rejection:** Sudden abnormal readings are ignored to prevent false detection.

These techniques enhance stability and ensure accurate object detection during operation.

2.6.3 Visual Identification

The laser helps the user instantly see the object location without checking the serial monitor.

2.7 Summary

This chapter explained the theoretical foundations that support the project, including radar principles, ultrasonic sensing, laser aiming theory, microcontroller processing, and ground-level detection concepts. These concepts provide the necessary scientific basis for understanding the system design in the next chapters.

3. DESIGN

3.1. General Information

In this chapter, the design architecture, operational logic, and theoretical considerations of the Autonomous Ground-Level Scanning Vehicle with Ultrasonic Sensing and Laser Target Indication are presented. The system is based on the Time-of-Flight (ToF) principle, commonly used in sonar and radar technologies. Instead of electromagnetic waves, the proposed system uses ultrasonic waves to detect obstacles at ground level.

The design integrates mechanical, electrical, and embedded control subsystems to achieve autonomous movement, environmental scanning, and directional target indication using a laser mechanism.

3.2. Dimensioning

The dimensional specifications of the system were defined based on the mechanical design developed in SolidWorks 2023, ensuring accurate integration and positioning of all electronic and mechanical components. The CAD model was used to optimize the structural layout, achieve proper space utilization, and maintain overall mechanical stability shown in Figure 3.1. of the autonomous vehicle during operation

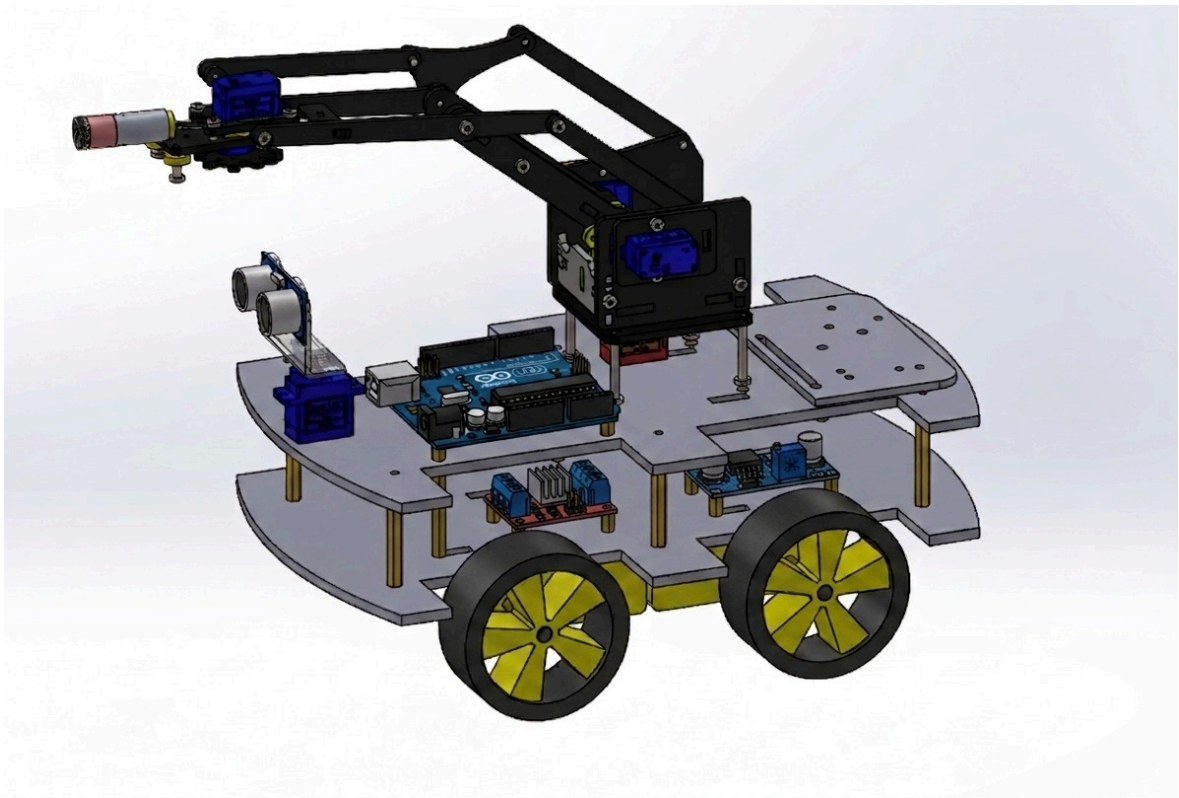


Figure 3.1. Solidworks assembly of the autonomous radar vehicle

The top view of the mechanical layout provides a clear visualization of component placement and spatial distribution across the chassis, as shown in Figure 3.2.

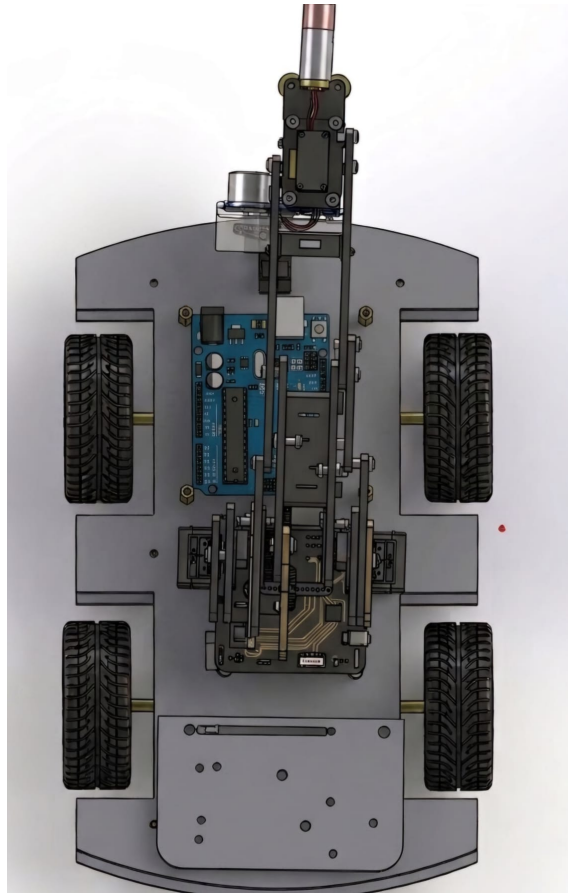


Figure 3.2. Top View of Mechanical Layout

3.2.1. Vehicle Chassis

The vehicle chassis serves as the primary structural platform that supports and integrates all mechanical and electronic components of the system. It is designed to ensure mechanical stability, proper weight distribution, and ease of component mounting while maintaining compact dimensions.

The chassis accommodates the following main components:

- Arduino Uno (central control unit)
- L298N motor driver module
- Breadboard for circuit connections
- Ultrasonic sensor with servo-based scanning mechanism
- Laser module and its mounting arm
- Battery pack (power supply unit)

The dimensional layout of the system was carefully designed to ensure proper alignment and spacing of all components, as illustrated in Figure 3.3.

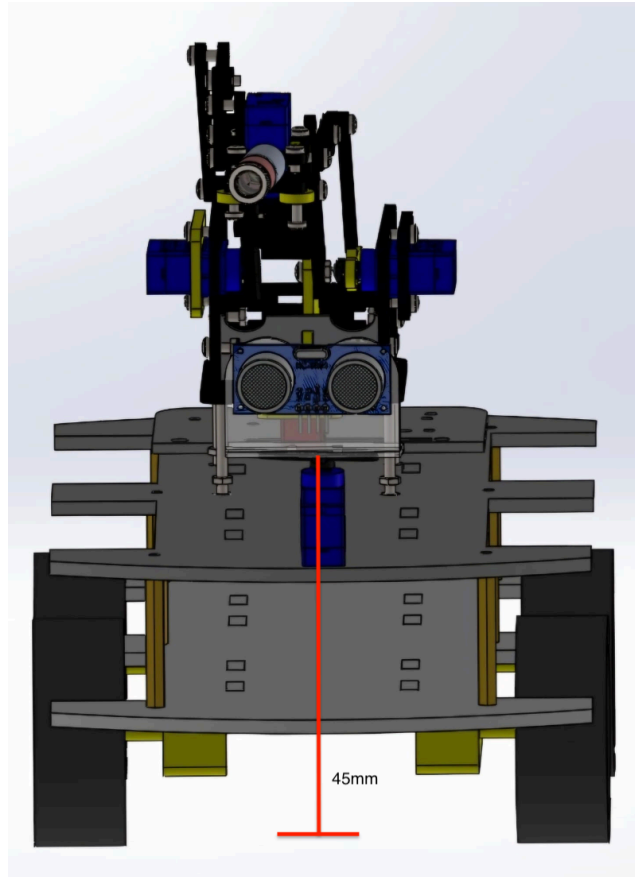


Figure 3.3. Dimensions View of Mechanical Layout

3.2.2. Sensor Mounting Height

The HC-SR04 ultrasonic sensor is mounted at a height of 40–50 mm to:

- Reduce ground reflection interference
- Maintain proper detection cone angle
- Ensure accurate obstacle distance measurement

3.2.3. Servo-Sensor Mechanism

The MG90S micro servo motor provides angular scanning between 0° and 180° . The ultrasonic sensor is mounted directly on the servo horn, allowing directional distance measurement.

This configuration enables the system to perform spatial scanning of the environment by changing the sensor orientation dynamically.

3.2.4. Laser Alignment

The laser module is mounted parallel to the ultrasonic sensor axis, ensuring that the laser points toward the same detected object direction. This alignment guarantees consistency between detection and visual indication.

The laser is positioned approximately 15 mm above the sensor centerline to minimize vertical parallax error and improve targeting accuracy.

3.2.5. Laser Arm Design

The laser targeting arm mechanism is designed to provide precise directional movement for object indication. The structural design and dimensions of this mechanism are shown in Figure 3.4.

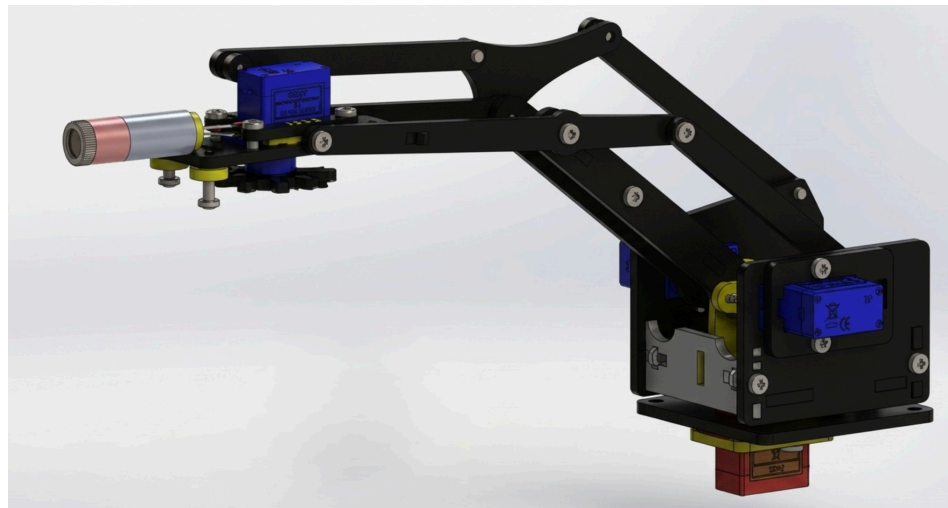


Figure 3.4. Dimensions View of laser arm design

3.2.6. Laser Targeting Arm Mechanism

The laser system is mounted on a dedicated lightweight mechanical arm integrated with the servo-based scanning system. This arm is responsible for directing the laser beam toward the detected object angle.

The laser arm is mechanically linked with the ultrasonic scanning servo system, ensuring synchronized movement between detection and indication.

Key features of the laser arm design:

- Mounted on a servo-controlled rotating platform
- Moves in synchronization with ultrasonic scanning angles
- Provides directional indication of detected obstacles
- Designed with lightweight material to minimize servo load
- Ensures stable alignment during rotation

This mechanism allows the system to visually highlight the exact direction of the detected object, improving system interpretability and demonstrating real-time tracking behavior.

3.3. System Components and Their Selections

In this project, each component was carefully selected based on functionality, cost, compatibility, and performance requirements. The system integrates sensing, processing, actuation, and user feedback to achieve autonomous object detection and targeting.

• Arduino Uno R3 (Processing Unit)

The **Arduino Uno** serves as the central control unit of the system. It was selected due to its simplicity, wide community support, and sufficient number of digital and analog I/O pins for handling multiple sensors and actuators.

The microcontroller is responsible for:

- Reading data from the ultrasonic and IR sensors
- Processing distance measurements
- Controlling motor movement and direction
- Driving the servo motor for radar scanning
- Coordinating the laser targeting system

Its ease of programming and compatibility with various modules make it ideal for rapid prototyping and educational projects.

• HC-SR04 Ultrasonic Sensor

The ultrasonic sensor is used as the primary distance measurement device in the system. It was chosen because it provides accurate distance readings within a practical range (~2 cm to 400 cm) and is not affected by lighting conditions.

This sensor plays a critical role in:

- Detecting obstacles in front of the vehicle
- Measuring the exact distance to objects
- Triggering the stop-and-scan behavior when an object is within 50 cm

Compared to IR sensors, ultrasonic sensing is more reliable for precise distance calculation, which is essential for radar functionality.

- **Servo Motor**

The servo motor is used to rotate the ultrasonic sensor, enabling scanning across multiple angles. It was selected because of its precise angular control (typically 0° to 180°).

Its function includes:

- Rotating the sensor to simulate radar scanning
- Allowing the system to detect the direction of obstacles
- Sending angle data to align the laser toward the detected object

Without the servo motor, the system would only detect objects in a single direction.

- **L298N Motor Driver**

DC motors are used to drive the movement of the car, while the L298N motor driver is used to control motor direction and speed.

They were selected because:

- DC motors provide sufficient torque for movement
- The L298N allows bidirectional control (forward/backward)
- It enables PWM speed control from the Arduino

This combination allows the vehicle to:

- Move forward during normal operation
- Stop when an obstacle is detected
- Change direction to avoid obstacles

- **Laser Module**

The laser module is used for visual targeting. It was selected to represent the “**Eye of Horus**” concept by pointing directly at detected objects.

Its purpose includes:

- Indicating the exact location of detected objects
- Enhancing visualization of the radar system
- Providing a clear and engaging output for demonstration

- **Infrared (IR) Sensors**

IR sensors are used as supplementary obstacle detection components. They were selected due to their fast response time and low cost.

Their role includes:

- Providing immediate detection of nearby obstacles

- Supporting the ultrasonic sensor in short-range detection
- Enhancing system safety by acting as a backup detection layer

Although IR sensors are less accurate in distance measurement, they are very effective for quick detection.

• **LCD Display**

The LCD is used to display system status messages. It was chosen to provide real-time feedback to the user.

Displayed information includes:

- System state (MOVING, STOPPED, SCANNING)
- Distance readings
- Detection alerts

This improves user interaction and debugging capability.

• **Power Supply (Battery + Buck Converter)**

The system uses a battery as the main power source, with a buck converter to regulate voltage levels.

This was necessary because:

- Different components require different voltages
- The Arduino and sensors operate at stable 5V
- Motors require higher current and sometimes higher voltage

The buck converter ensures:

- Stable operation
- Protection of sensitive components
- Efficient power distribution

• **Buzzer Module**

The buzzer is used as an audible alert component in the system. It was selected due to its simplicity, low cost, and ease of integration with the microcontroller.

Its main purpose is to:

- Provide an immediate sound indication when an obstacle is detected
- Enhance system awareness without relying only on visual outputs (LCD and laser)
- Assist in debugging and testing by signaling detection events

- The buzzer is activated when the detected distance is below the predefined threshold (30 cm) and deactivated otherwise. This ensures that the user receives real-time auditory feedback about nearby obstacles.

- **HC-05 Bluetooth Module**

The HC-05 Bluetooth module was integrated into the system to provide wireless communication between the vehicle and a smartphone application. This allows real-time manual control and enables smooth switching between autonomous and manual operating modes.

3.4. Applied Methods

The system operates using a Hybrid Autonomous Scanning Method, which combines continuous vehicle motion with periodic environmental scanning and decision-making. This approach ensures both efficient navigation and accurate object detection.

3.4.1. Vehicle Movement

The mobile platform moves forward at a constant or user-defined speed under normal conditions. During this phase, the system continuously monitors sensor inputs without interrupting motion.

This method enables:

- Efficient area coverage
- Continuous operation without unnecessary(stopping)
- Real-time responsiveness to environmental changes

3.4.2. Servo-Based Radar Scanning

The radar functionality is achieved by rotating the ultrasonic sensor using a servo motor across a range of angles (0° to 180°).

At each scanning step:

- The servo motor moves to a specific angle θ
- The microcontroller sends a **10 μ s trigger pulse** to the ultrasonic sensor
- The echo signal duration is measured using the `pulseIn()` function

This method allows the system to:

- Scan multiple directions
- Determine the angular position of detected objects
- Simulate radar-like behavior

3.4.3. Object Detection Logic

The system applies a threshold-based decision mechanism to determine the presence of obstacles.

- A predefined threshold distance (e.g., **30 cm**) is used
- The measured distance is continuously compared against this value

Decision Logic:

If (*Distance < Threshold*):

Laser module is activated → Buzzer is turned ON → LCD displays the detected angle and distance → Else → Laser and buzzer remain OFF

3.4.4. Continuous Loop

The scanning and detection process is executed in a continuous loop with a refresh interval of approximately **100–150 ms**.

This timing is carefully selected to:

- Reduce false readings caused by ultrasonic reflections
- Maintain system stability
- Ensure smooth and real-time operation

3.5. Software Used

The software development for this project was carried out using the Arduino IDE, which provides an integrated platform for writing, compiling, and uploading code to microcontroller-based systems.

The Arduino IDE was selected due to its:

- User-friendly interface
- Extensive library support for sensors and actuators
- Compatibility with a wide range of hardware modules
- Strong community support, which facilitates debugging and development

The programming language used is based on **C/C++**, allowing efficient low-level control of hardware components while maintaining code simplicity.

Development Approach

The software follows a **structured and iterative development approach**, where individual modules (such as sensing, actuation, and display) were developed and tested separately before system integration.

The final program is organized into two main functional sections:

- **Initialization (setup()):** Responsible for configuring hardware components and setting initial states
- **Main Loop (loop()):** Executes continuous system operations including scanning, detection, and response

The embedded software for the system was developed and tested using the Arduino Integrated Development Environment (IDE). The platform provides tools for writing, compiling, and debugging the control code that governs sensor interaction, motor control, and system logic. An example of the developed program, including servo configuration and pin assignments, is shown in Figure 3.5.

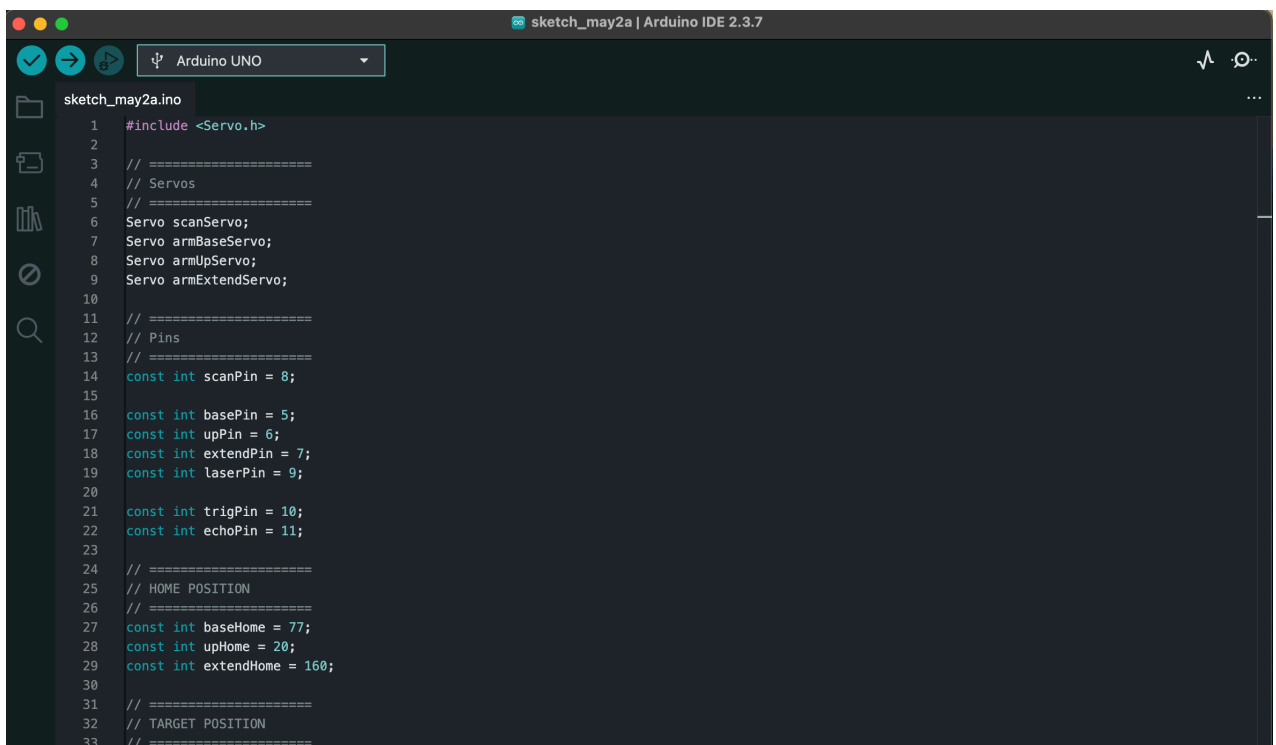


Figure 3.5. Arduino IDE interface showing the implemented control code

As shown in Figure 3.5, the program includes the definition of servo objects and pin configurations required for controlling the robotic arm and sensing components. The code structure demonstrates how multiple actuators and sensors are initialized and managed within the Arduino environment. This implementation ensures synchronized operation between scanning, detection, and targeting mechanisms.

3.6. Component List and Economic Analysis

The prices listed are based on current market averages from local electronics suppliers (e.g., Robotistan, Direnc.net). note: the prices is able to change

Table 3.1 Component List

Component Name	Purpose of use	Unit Price (TL)	Quantity	Price (TL)
Arduino Uno R3	Processing unit	300	1	300
HC-SR04 Sensor	Distance measurement	150	1	150
MG90 Servo	Scanning mechanism	200	4	800
DC Motor + Wheels	Vehicle movement	1500	1	1500
L298N Motor Driver	Motor control	200	1	200
Buzzer	Audible warning	40	1	40
32x4 LCD	Display output	350	1	350
Laser Module	Visual marking	120	1	120
Buck converters (LM2596)	Voltage regulator	150	6	900
Breadboard	Prototyping	80	1	80
HC-05 Bluetooth Module	Wireless manual control	250	1	250
Battery	Power supply	400	1	400
TOTAL				5250

Economical Analysis

The total estimated cost of the project is approximately 5250 TL. The selected components were chosen based on affordability, availability, and compatibility with the system requirements. Although some components increased the overall cost, the project still provides an economically efficient and reliable solution suitable for educational and small-scale robotic applications.

3.7. Legal Aspects

The proposed system has been designed in compliance with relevant international and national regulations to ensure safe operation, environmental responsibility, and legal conformity.

3.7.1. Laser Safety (IEC 60825-1)

The laser module used in the system complies with the **IEC 60825-1 laser safety standard**, which classifies laser products according to their potential hazard level.

- The implemented laser is classified as **Class 1 or Class 2**, which is considered safe for educational and demonstration purposes
- The laser is operated at low power, minimizing any risk of biological damage
- It is mechanically fixed at a controlled downward angle, preventing direct eye exposure during normal operation

These measures ensure that the system can be safely used in laboratory and academic environments.

3.7.2. Waste Electrical and Electronic Equipment (WEEE) Compliance

The system components fall under the scope of the **EU Waste Electrical and Electronic Equipment (WEEE) Directive**.

To comply with environmental regulations:

- Electronic components will be disposed of at certified electronic waste recycling facilities
- Reusable components may be recovered for future projects
- Hazardous electronic waste is separated and handled according to environmental safety standards

This approach minimizes environmental impact and supports sustainable engineering practices.

3.7.3. National Electronics Regulations

The system operates within standard low-voltage electronic limits and does not require special licensing.

- The system is powered within the **7–12V low-voltage range**, which is considered safe for educational prototypes
- The system does not include any radio frequency (RF) transmission modules

4. SIMULATION STUDIES

4.1. General Information

This chapter describes the simulation-based implementation of the **Autonomous Ground-Level Scanning Vehicle with Ultrasonic Sensing and Laser Target Indication**. The simulation phase is used as a verification stage alongside the hardware development process in order to validate system behavior, embedded control logic, and component interactions before final integration.

The simulation is carried out using **Tinkercad** Circuits to analyze and confirm the functionality of the ultrasonic scanning mechanism, servo motor movement, laser activation, buzzer response, LCD display output, and motor control logic under different operating conditions.

The simulation environment provides a safe, flexible, and cost-effective platform for testing circuit connections, embedded software logic, and real-time system responses before and during hardware implementation.

4.2 Simulation Environment

4.2.1. Tinkercad Circuits

Autodesk Tinkercad Circuits is used as the primary simulation tool due to its real-time Arduino execution capability, ease of use, and support for essential electronic components such as ultrasonic sensors, servo motors, LCD modules, LEDs, and buzzers.

The platform allows visualization and validation of circuit design, as well as dynamic testing of system behavior by adjusting simulated object distance and observing system response.

In this project, **Tinkercad** is used to verify the system logic and control behavior, including:

- Servo scanning motion
- Ultrasonic distance measurement
- LCD information display
- Buzzer activation for obstacle detection
- Laser module control logic
- Motor driver signal behavior

While **Tinkercad** does not fully replicate real-world physical effects such as noise, mechanical inertia, or sensor inaccuracies, it provides a reliable environment for validating the embedded control algorithm and system-level functionality prior to full hardware deployment.

4.3. Circuit Design and Wiring

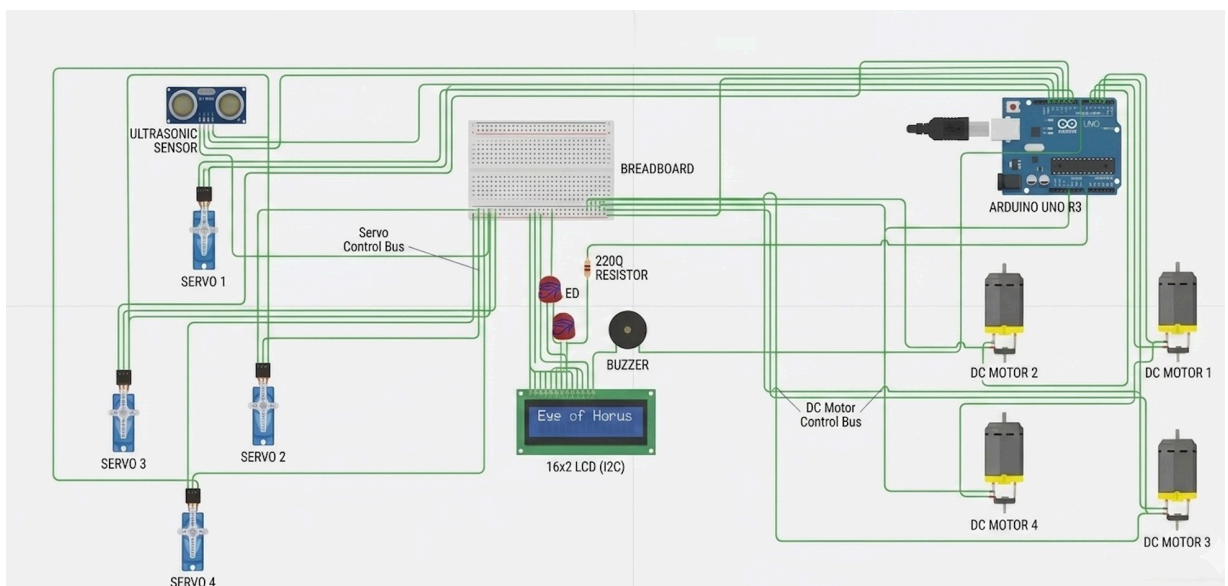


Figure 4.1. Full Wiring Diagram of the Radar System

From Figure 4.1, it can be observed that the system follows a centralized control architecture where the Arduino Uno manages all input and output operations. The wiring configuration ensures proper synchronization between sensing components (ultrasonic sensor), actuation elements (servo motors and DC motors), and feedback modules (LCD, buzzer, and laser). Special attention is given to signal routing and power distribution to avoid interference between high-current motor lines and low-power sensor circuits, thereby improving system stability and reliability.

A detailed view of the ultrasonic sensor connections and its interaction with other components is presented in Figure 4.2, highlighting the signal flow and wiring configuration.

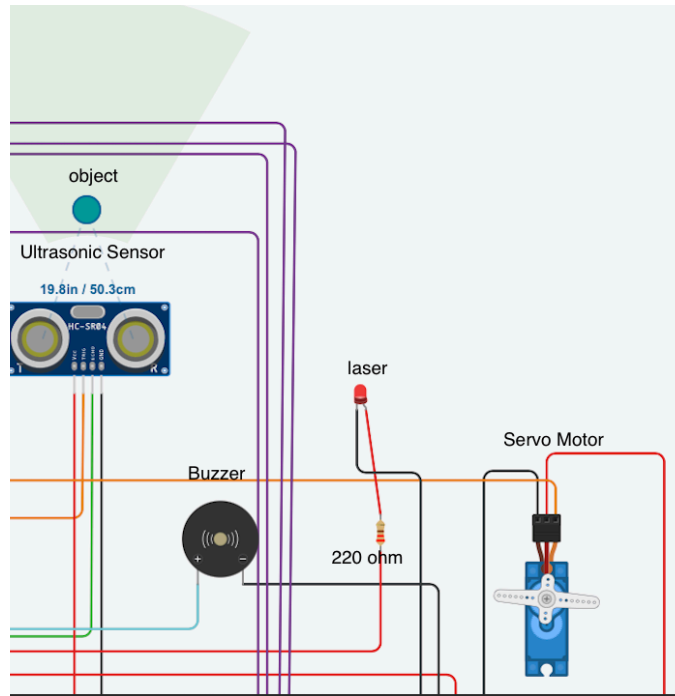


Figure 4.2. Close-Up View of Ultrasonic Sensor

4.4. Simulation Setup and Initialization

The initialization of the system is verified through the LCD display, which shows the startup message indicating successful communication with the microcontroller, as illustrated in Figure 4.3.

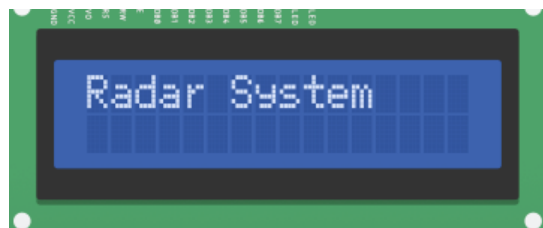


Figure 4.3. LCD Initialization Test

Initial LCD test confirming successful communication with the Arduino

At system startup:

- The Arduino initializes all input and output pins
- The LCD displays system status (“**Radar System**”)

This step ensures that all hardware components are properly initialized before system operation begins.

4.5. Simulation Workflow

The real-time behavior of the system during operation, including ultrasonic detection, servo movement, and system response, is demonstrated in Figure 4.4.

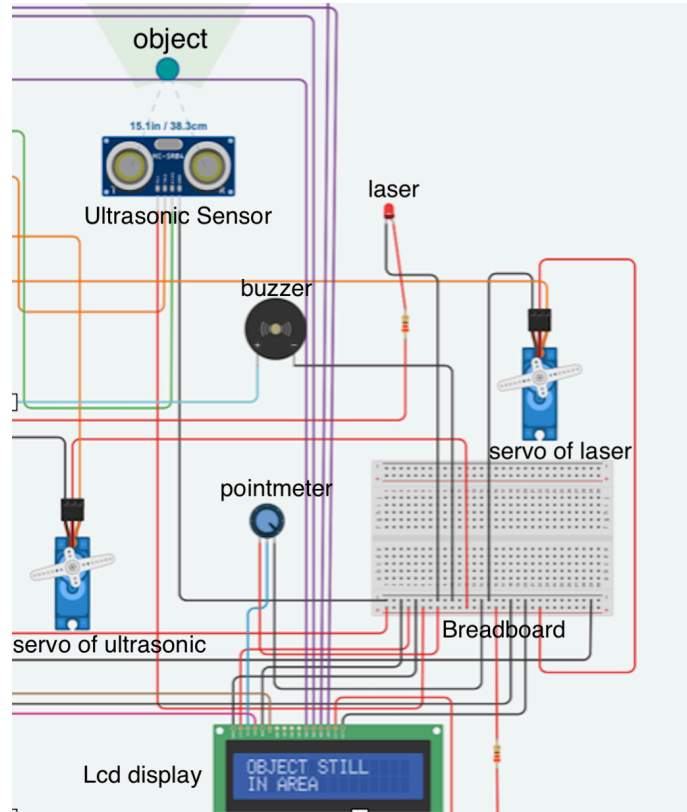


Figure 4.4. Real-time behavior of the autonomous scanning system

4.5.1 Continuous Movement Mode

The vehicle operates in an autonomous forward movement state while continuously monitoring the environment using the ultrasonic sensor.

Movement control is based on sensor feedback rather than fixed timing, allowing dynamic response to obstacles.

4.5.2 Scanning Process

The servo motor performs angular scanning across a defined range (0° to 180°). At each angle:

- The ultrasonic sensor transmits a trigger pulse
- The echo signal is received and processed
- Distance is calculated using time-of-flight principle
- The system maps distance values to angular positions

This process enables directional detection of surrounding obstacles.

4.5.3 Object Detection

When an object is detected within the defined threshold (≤ 50 cm):

- The vehicle immediately stops
- The system records the detected angle
- The laser module is activated
- The servo aligns the laser toward the detected object
- The buzzer is activated as an alert signal

If no object is detected within range:

- The system continues forward movement
- Laser and buzzer remain OFF
- Scanning continues continuously

4.5.4 Manual Movement Mode

- Forward/backward/left/right control
- Bluetooth commands
- Switching between modes

4.5.5 Real-Time Control Loop

The system operates in a continuous loop where scanning, movement, and decision-making processes are dynamically updated based on real-time sensor input. This loop-based structure allows the microcontroller to continuously read data from the ultrasonic sensor and other inputs, process the information, and generate appropriate control signals for actuators such as motors, servos, and output modules.

During each iteration of the loop, the system performs distance measurement, evaluates detection conditions, and determines the required response, such as continuing movement or activating the laser and buzzer. This ensures that the system maintains constant interaction with its environment.

As a result, the control loop provides continuous environmental awareness, fast response time, and reliable operation under changing conditions, enabling the system to react immediately to obstacles and maintain stable autonomous behavior.

TABLE 4.1. Simulation Scenarios and Outputs

Scenario	System Behavior	Result
No Object Present	Full 0–180° scan	Laser OFF, buzzer OFF, car moves
Object at 70°	Scan stops at 70°	Laser ON, buzzer ON, lock mode
Object Stays in Area	Servo fixed at 70°	Car stays stopped
Object Leaves	Distance > threshold	Laser OFF, buzzer OFF, resume movement
Far Object (>50 cm)	Out of threshold	Normal scanning

TABLE 4.2. LCD Display States During Simulation

System State	LCD Line 1	LCD Line 2
Movement Phase	“Eye Of Horus”	“Moving”
Scanning (No Object)	“Ang:XX°”	“No Object”
Object Locked	“OBJECT LOCKED”	“Ang:XX° D:XXcm”
Object Still	“OBJECT STILL”	“IN AREA”
Area Cleared	“AREA CLEARED”	“Resume Move”

TABLE 4.3. Object Detection and Laser Alignment Data

Angle (°)	Distance (cm)	Detection Status	Laser Angle
30°	120	No object	OFF
60°	55	Near threshold	OFF
70°	24	Detected	70°
70° (locked)	22	Still detected	70°
70° (after clear)	120	Cleared	OFF

5. EXPERIMENTAL STUDIES

5.1. General Information

This chapter presents the physical implementation of **the Autonomous Ground-Level Scanning Vehicle with Ultrasonic Sensing and Laser Target Indication**. The prototype was developed based on the design and simulation results presented in previous chapters.

The objective of this phase is to integrate all mechanical, electrical, and embedded components into a functional system and evaluate its real-world performance. The prototype demonstrates autonomous movement, obstacle detection, directional laser indication, and real-time system feedback.

5.2 System Integration Overview

The hardware components introduced in the design phase were physically assembled and integrated into a complete operational system. Rather than focusing on component selection, this stage emphasizes how sensing, control, and actuation subsystems interact in real-time.

The prototype combines multiple subsystems, including motion control, distance sensing, short-range detection, and a robotic arm mechanism for laser targeting. These subsystems operate together under the control of the Arduino-based embedded system, forming a closed-loop autonomous platform.

The integration process required careful arrangement of components across the chassis, proper routing of wiring, and synchronization between sensors and actuators to ensure stable and reliable system behavior.

5.3 Mechanical Assembly

The mechanical structure of the prototype is based on a multi-layer chassis design that provides stability, modularity, and efficient use of space. The system is built on a four-wheel drive (4WD) platform, ensuring balanced load distribution and reliable movement.

The chassis is divided into 3 functional layers:

Lower layer: DC motors, wheels, and main power components

Middle layer: control electronics, including the bucks and motor driver

Upper layer: sensing , actuation systems and Arduino

A robotic arm mechanism is mounted on the upper layer of the vehicle. This arm is driven by servo motors and is responsible for positioning the laser module toward detected objects. The structure of the arm allows controlled movement and stable operation during targeting.

The ultrasonic sensor is positioned at the front of the vehicle to enable forward obstacle detection, while multiple IR sensors are mounted near the front edges to enhance short-range sensing and improve environmental awareness.

The overall mechanical design ensures stability during movement, proper alignment of sensing components, and secure mounting of all subsystems as shown in Figure 5.1.

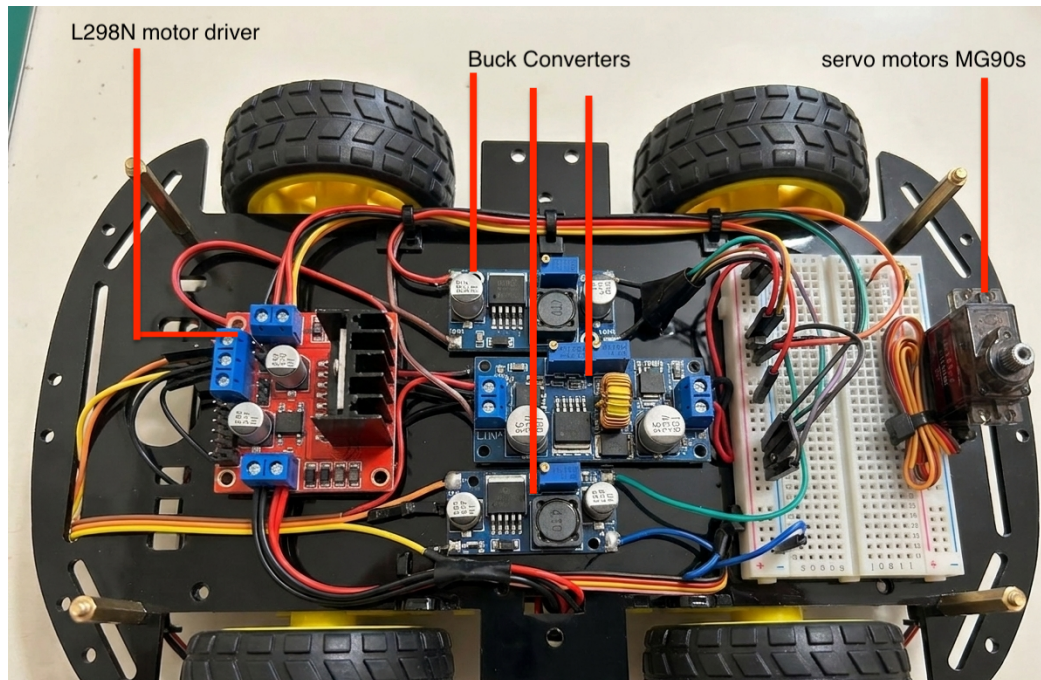


Figure 5.1. Intermediate stage of prototype development and component placement

From Figure 5.1, the spatial distribution of the core power and control modules. Centrally located are the DC-DC buck converters for voltage regulation, flanked by the L298N motor driver (left) and a prototyping breadboard (right) and servo motor. This configuration was chosen to minimize electromagnetic interference and optimize wire routing between the power source and the four-wheel drive system.

5.4 Electrical Connections

The electrical system was implemented through organized wiring distributed across the chassis layers. A rechargeable battery pack supplies power to the system, while a voltage regulation module ensures stable voltage levels for different components.

The electrical integration includes connections between:

- The Arduino and motor driver for motion control
- The ultrasonic sensor for distance measurement
- Servo motors for robotic arm movement
- Output components such as the laser module and buzzer

Special attention was given to grounding and power distribution to reduce electrical noise and ensure reliable signal transmission. Wiring was arranged to avoid interference between high-current motor lines and low-signal sensor connections. The bottom view of the robotic chassis highlights the power supply system and drive mechanism, including battery placement and motor connections, as illustrated in Figure 5.2.

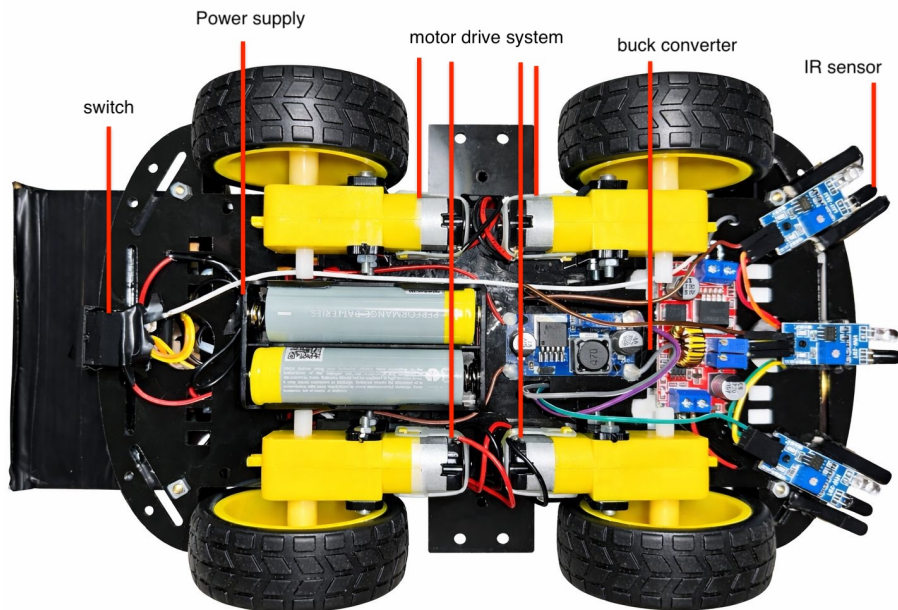


Figure 5.2. Bottom view of the robotic chassis showing power and drive system

From Figure 5.2, battery power supply, the four-motor drive system, and the secondary power regulation modules. The layout demonstrates the routing of high-current wiring from the batteries to the motor controllers situated on the upper deck.

5.5 Embedded System Implementation

The system was programmed using Arduino C/C++ to implement real-time control logic. The embedded software integrates multiple functionalities, including:

- Continuous distance measurement using ultrasonic sensing
- Processing of IR sensor inputs for short-range detection
- Control of DC motors for autonomous movement
- Servo motor control for robotic arm positioning
- Decision-making based on distance threshold conditions
- Activation of laser and buzzer upon obstacle detection

5.6 Prototype Operation

During operation, the vehicle moves autonomously while continuously monitoring its surroundings. The ultrasonic sensor detects objects in front of the vehicle, while IR sensors provide additional short-range detection capability as shown in Figure 5.3.

When an object is detected within the predefined threshold distance:

- The vehicle stops immediately
- The robotic arm adjusts its orientation toward the detected object
- The laser module is activated to indicate the object's position
- The buzzer generates an alert signal

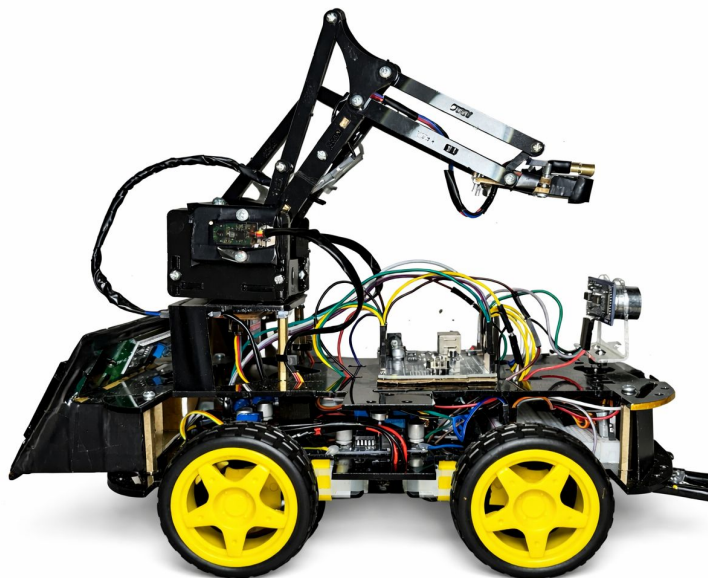


Figure 5.3. Lateral View of the Integrated Prototype.

5.7 Results and Observations

The prototype was tested under controlled conditions to evaluate system performance. The following results were observed and the final prototype is presented in Figure 5.4.

- Reliable detection of obstacles using ultrasonic sensing
- Stable and balanced movement using the 4WD system
- Effective operation of the robotic arm for directional targeting
- Successful activation of laser and buzzer upon detection
- Improved environmental awareness due to IR sensors

Some limitations were also identified:

- Ultrasonic sensor accuracy depends on surface properties and angle
- Mechanical vibrations may introduce minor measurement variations
- Power fluctuations can affect motor performance
- Servo speed limits the response time of the robotic arm

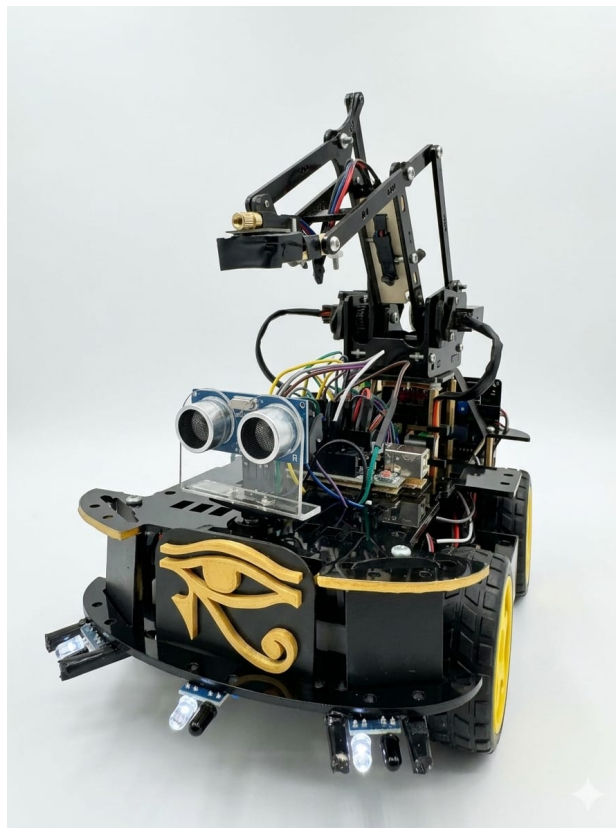


Figure 5.4. Final Integrated Autonomous Robotic System "Eye of Horus."

From Figure 5.4. the fully assembled prototype, integrating the mobile chassis, robotic arm, and sensor suite (Ultrasonic and IR) for autonomous navigation and task execution.

6. RESULTS

6.1 Introduction

This chapter presents the results obtained from both simulation and experimental studies of the proposed autonomous scanning vehicle system. The purpose of this chapter is to evaluate whether the system meets the intended design objectives, including accurate obstacle detection, directional sensing, and real-time response. The results are presented in the form of numerical data, tables, and graphical analysis. Simulation results are first discussed under ideal conditions, followed by experimental results obtained from the physical prototype. A comparison between both results is also provided to system performance and reliability. And the system during the object detection as shown in Figure 6.1.

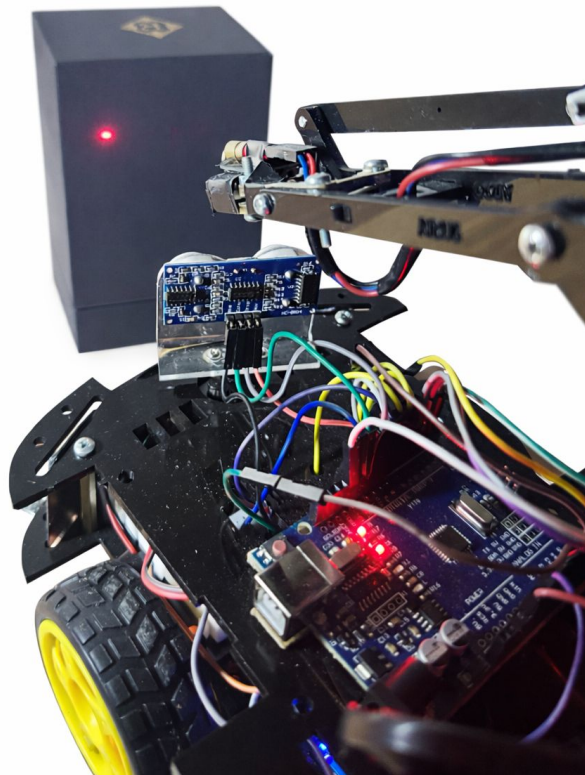


Figure 6.1. Prototype during operation showing obstacle detection and laser targeting

The system behavior during obstacle detection is illustrated in Figure 6.1. The laser module is directed toward the detected object, confirming correct alignment between the ultrasonic sensing system and the robotic arm mechanism. This visual result supports the data obtained from experimental measurements.

6.2 Simulation Results

The simulation results were obtained using **Tinkercad** Circuits to analyze system behavior under controlled conditions. The ultrasonic sensing system was tested at different distances, and the measured values were compared with expected values.

TABLE 6.1 Simulated Distance Measurements in (Tinkercad)

Test No.	Actual Distance (cm)	Measured Distance (cm)	Error Percentage (%)
1	20	20	0.0%
2	30	30	0.0%
3	40	40	0.0%
4	50	50	0.0%
5	60	60	0.0%

The results show that the system provides highly accurate distance measurements under simulation conditions, with negligible deviation. This confirms that the ultrasonic sensing logic and control algorithm are correctly implemented.

The simulation also verified that the detection threshold was properly applied, allowing the system to identify objects within the predefined range in Figure 6.2.

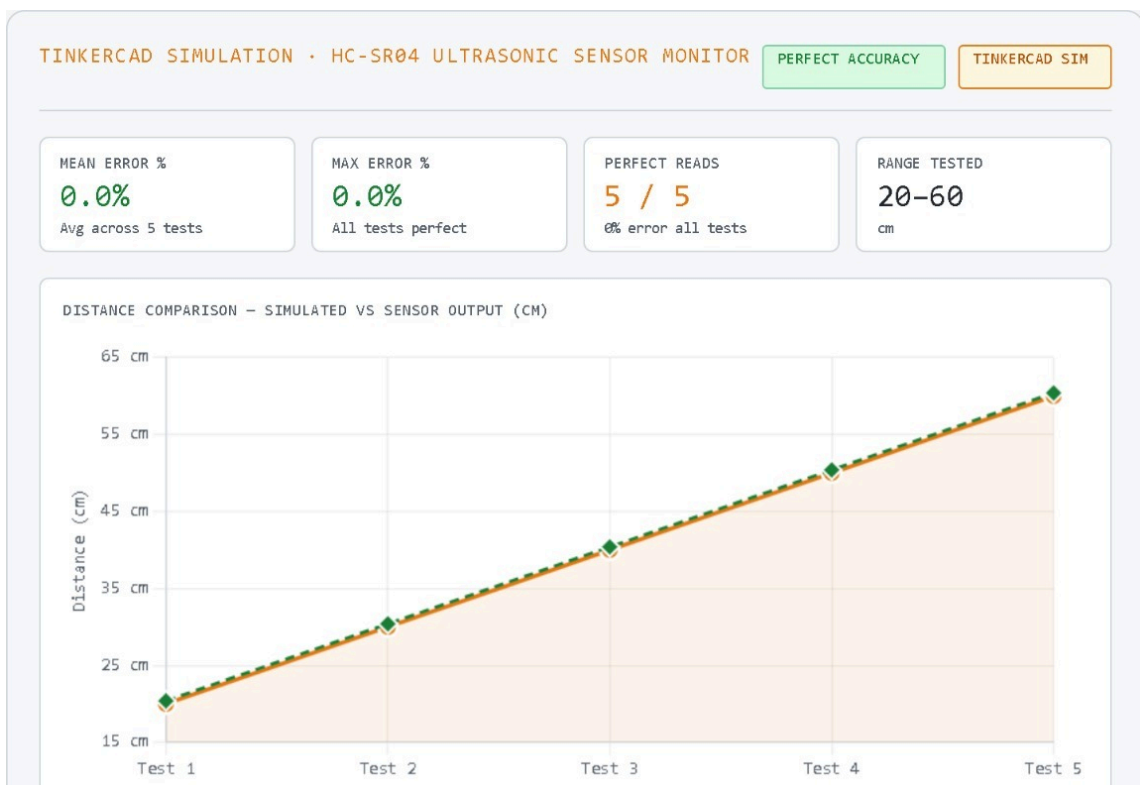


Figure 6.2. Simulated Distance vs Measured Distance (Tinkercad)

6.3. Experimental Results

Experimental results were obtained by testing the physical prototype under real operating conditions. Measurements were recorded

6.3.1 Ultrasonic Sensor Accuracy and Linearity

The distance sensing accuracy was tested on the physical prototype. Minor deviations were expected due to atmospheric noise and signal reflection.

Table 6.2 Experimental Sensor Accuracy Data

Actual Distance (cm)	Measured Distance (cm)	Error Percentage (%)
20	20	0.0%
30	31	3.3%
40	40	0.0%
50	52	4.0%
60	59	1.6%

As shown in Figure 6.3, the experimental data follows the ideal reference line closely. The maximum error of 4.0% is well within acceptable limits for autonomous navigation and target acquisition.

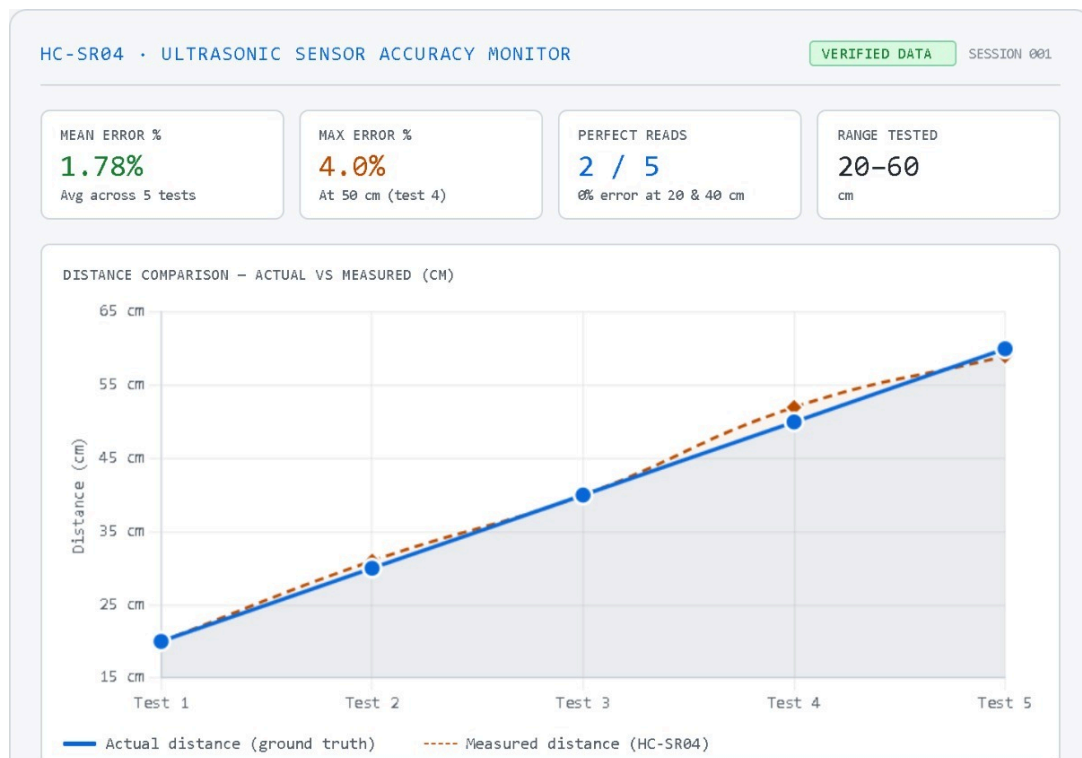


Figure 6.3. Experimental Results: Actual vs. Measured Distance.

6.3.2 Operational Success Rate and Functional Validation

The robot's ability to execute its "Search and Engage" logic was tested over 15 trials. This test validates the coordination between the ultrasonic sensor, the robotic arm, and the laser.

Table 6.3: System Functional Success Rate (15 Trials)

Functional Parameter	Successes	Failures	Success Rate (%)
Obstacle Detection	15	0	%100
Laser Actuation	13	2	86.6%
LCD Status Update	15	0	%100

While detection and UI reporting achieved 100% success, the laser alignment experienced a 13.4% failure rate. This was observed when the robot stopped abruptly, causing mechanical inertia in the arm to shift the laser slightly off-target in Figure 6.4.

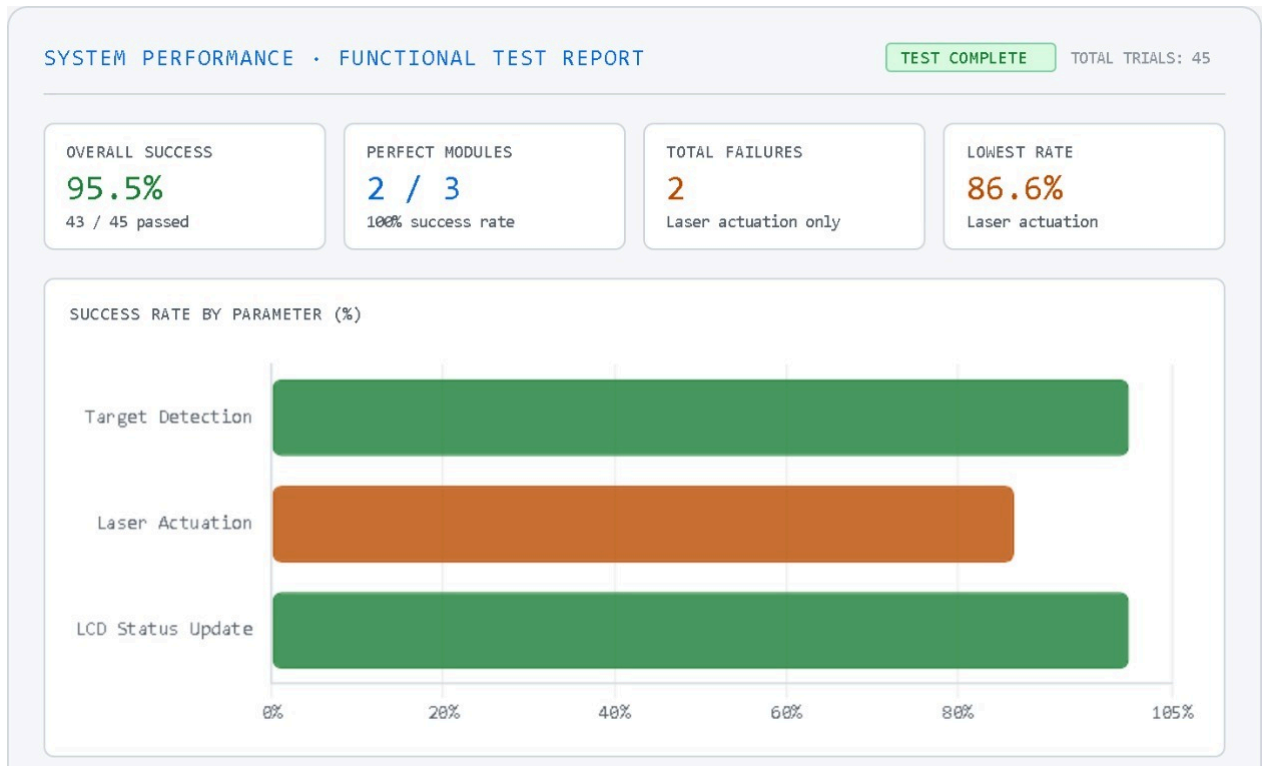


Figure 6.4. System Functional Success Rate Chart.

6.4 Comparison of Simulation and Experimental Results

A comparison between simulation and experimental results shows that:

- Simulation results exhibit ideal and highly accurate behavior
- Experimental results show minor deviations due to real-world factors
- The system performance remains consistent with design expectations

The differences are mainly caused by:

- sensor noise
- surface reflection variations
- mechanical vibrations

Despite these differences, the overall system behavior remains reliable and meets the intended objectives.

6.5 Discussion

The results confirm that the proposed system successfully achieves its main objectives, including obstacle detection, directional indication, and autonomous response. Both simulation and experimental outcomes demonstrate that the system is capable of identifying obstacles within the defined range and reacting appropriately in real time.

The system also shows effective integration between the sensing, control, and actuation subsystems. The ultrasonic sensor provides reliable distance measurements, while the servo mechanism ensures accurate directional scanning. The control logic successfully processes the input data and triggers the correct system response without noticeable delay.

Although minor deviations were observed between simulation and experimental results, these differences are expected due to real-world factors such as sensor noise, mechanical tolerances, and environmental interference. However, these variations do not significantly affect the overall system performance and remain within acceptable limits for this type of embedded application.

Overall, the results indicate that the system is stable, responsive, and suitable for autonomous scanning tasks, confirming the validity of the proposed design approach.

7. CONCLUSIONS

This project presented the design, simulation, and implementation of an Autonomous Ground-Level Scanning Vehicle with Ultrasonic Sensing and Laser Target Indication. The system integrates sensing, control, and actuation mechanisms to achieve real-time obstacle detection and directional response.

The simulation studies confirmed that the proposed control logic operates correctly under ideal conditions, providing accurate distance measurement and proper system behavior. The experimental results obtained from the physical prototype demonstrated that the system performs reliably in real-world conditions, with only minor deviations due to environmental factors and hardware limitations.

The prototype successfully achieved its main objectives, including autonomous movement, obstacle detection within a predefined threshold, directional laser targeting, and alert generation. The integration of ultrasonic sensing with a servo-controlled mechanism enabled effective scanning and object localization, while the robotic arm provided accurate visual indication of detected obstacles.

Overall, the results validate the effectiveness of the proposed system design. The combination of simulation and experimental validation confirms that the system meets the intended goals and can serve as a functional autonomous detection and indication platform.

7.1 Future Work

Although the system demonstrates reliable performance, several improvements can be implemented to enhance its capabilities and extend its applications:

- Integration of **line-following or path planning algorithms** to transform the system into a fully functional autonomous guided vehicle
- Use of **higher-precision sensors** (e.g., LiDAR or advanced ultrasonic modules) to improve detection accuracy
- Enhancement of the robotic arm mechanism for faster and more precise targeting
- Addition of a **camera system** for visual object recognition and advanced tracking

These improvements would increase system performance, expand functionality, and enable more advanced real-world applications.

8. REFERENCES

- [1] R. C. Dorf, R. H. Bishop, “Modern Control Systems,” Pearson, 2017.
- [2] A. S. Grove, “Sensors and Signal Conditioning,” Wiley, 3rd Ed., 2015.
- [3] J. G. Webster, “The Measurement, Instrumentation, and Sensors Handbook,” CRC Press, 1999.
- [4] S. K. Sahoo, P. K. Sahu, “Short-Range Ultrasonic Distance Measurement for Robotics Applications,” *International Journal of Advanced Engineering Research and Science*, vol. 5, no. 2, pp. 112–118, 2018.
- [5] E. K. Koutroulis, “Design of a Mobile Robot for Autonomous Ground Navigation Using Ultrasonic Sensors,” *IEEE Sensors Journal*, vol. 19, no. 3, pp. 1045–1053, 2019.
- [6] M. R. Shakarami, F. Faghihi, “Performance Evaluation of Ultrasonic Sensors in Ground-Level Object Detection,” *Journal of Robotics and Control*, vol. 7, no. 4, pp. 254–260, 2020.
- [7] Y. Zheng, T. Tamaki, “Laser-Based Visual Position Indication for Indoor Mobile Robots,” in *Proc. IEEE Int. Conf. Robotics and Automation*, Montreal, Canada, 2019, pp. 4771–4776.
- [8] A. Rahman, K. Ahmed, “Ultrasonic-Based Autonomous Navigation for Low-Cost Educational Robots,” in *Proc. International Symposium on Robotics Systems*, Istanbul, Turkey, 2021, pp. 55–60.
- [9] H. Yılmaz, “Design and Implementation of an Ultrasonic Obstacle Detection System,” Master’s Thesis, Istanbul Technical University, Electrical and Electronics Engineering Department, 2020.
- [10] F. A. Karim, “Laser-Assisted Object Indication for Mobile Robots,” Graduation Project, Middle East Technical University, Mechanical Engineering Department, 2021.
- [11] Arduino Official Website, <https://www.arduino.cc/>
- [12] HC-SR04 Ultrasonic Sensor Technical Documentation, <https://www.makerguides.com/hcsr04-arduino-tutorial/>
- [13] “MG90S Metal Gear Micro Servo Motor Datasheet,” TowerPro. [Online]. Available: <https://components101.com/motors/mg90s-metal-gear-servo-motor>
- [14] Tinkercad Learning Resources, <https://www.tinkercad.com/learn>
- [15] SparkFun Laser Module Product Guide, <https://www.sparkfun.com/products/retired/10549>
- [16] “L298N Dual H-Bridge Motor Driver,” STMicroelectronics. [Online]. Available: <https://www.st.com/resource/en/datasheet/l298.pdf>
- [17] HC-05 Bluetooth Module Technical Documentation, <https://components101.com/wireless/hc-05-bluetooth-module>



IEEE Code of Ethics



We, the members of the IEEE, in recognition of the importance of our technologies in affecting the quality of life throughout the world, and in accepting a personal obligation to our profession, its members and the communities we serve, do hereby commit ourselves to the highest ethical and professional conduct and agree:

1. to accept responsibility in making engineering decisions consistent with the safety, health and welfare of the public, and to disclose promptly factors that might endanger the public or the environment;
2. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;
3. to be honest and realistic in stating claims or estimates based on available data;
4. to reject bribery in all its forms;
5. to improve the understanding of technology, its appropriate application, and potential consequences;
6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;
7. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, and to credit properly the contributions of others;
8. to treat fairly all persons regardless of such factors as race, religion, gender, disability, age, or national origin;
9. to avoid injuring others, their property, reputation, or employment by false or malicious action;
10. to assist colleagues and co-workers in their professional development and to support them in following this code of ethics.

Approved by the IEEE Board of Directors
August 1990

DECLARATION

Ethical Behavior and Professional Honesty

Fair and accurate assessment are key factors that must be at the core of education. Any act of a student that would cause unfair advantage to himself/herself or another student is considered an academic dishonesty. It is also academic dishonesty to copy someone else's work and make it look like his/her own. This is called theft.

Allowing someone else to copy your work and make it look like his/her own is also fraud. It is both wrong and unethical to provide all kinds of unfair profits and advantages. It is also wrong to prepare the assignments, projects or laboratory reports of the other students. Doing this is not ethical, neither. Likewise, having his/her own homework, project or laboratory report being prepared and written by someone else is also a complete deception and immorality.

Using another student's answers, homework or reports without his/her knowledge is nothing but stealing that student's labor and this is called theft. It is the same thing to pass the course by cheating in exams.

We should not forget that those who cannot learn the truth when they are students continue these mistakes after graduation and become one of the characters we complain about in the society.

We expect our students and alumni to have ethical and humanly behaviors during both their professional business and social lives. For this, we want you to promise us that you will obey the ethical rules by signing this form.

Let us fight with unethical behavior together.


Prof. Dr. İsmail H. ALTAŞ
Department Chair

I believe what is written above is correct within the framework of ethical rules. I promise to behave ethically during and after my education.

Student No	Student Name and Surname	Date	Signature
210202885	AHMED MOHAMED IBRAHIM MOSTAFA	18.12.2025	
210202967	AHMED MAHMOUD SOLIMAN SOLIMAN	18.12.2025	
210202996	AMR AYMAN MOHAMED ELMOATASEMBELLAH	18.12.2025	
210202943	ABDALLA ELSAYED HOSSNI MOHAMED AHMED	18.12.2025	
220202019	EBRU KARCI	18.12.2025	

Each student must fill out this form with a blue pen, sign it, and add it to Graduation Project file as an Appendix

GRADUATION PROJECT RESTRICTIONS FORM

Explain the following constraints that may be encountered at universal and societal levels from project design to implementation stages.

Restrictions	Explanations
Design restrictions	The project design was limited by the size and weight of the robotic vehicle, power consumption, and Arduino processing capability while maintaining stable system performance.
Limitations in Prototype production	The prototype implementation was affected by component availability, motor vibration, sensor accuracy, and limited manufacturing resources during assembly.
Application restrictions	The system requires a stable power supply, regular maintenance, and careful operation to ensure safe and reliable performance during real-world usage.
Social influence and restrictions	The project contributes to robotics education and smart monitoring systems, but it must be used responsibly to avoid privacy misuse and unsafe applications.
Health restrictions	The laser module must not be directed toward human eyes, and moving mechanical parts should be handled carefully during testing and operation.
Environmental impact and restrictions	Electronic waste and battery disposal must follow environmental regulations to avoid pollution. Excessive energy consumption and improper disposal of damaged electronic components may negatively affect the environment.
Safety restrictions	Proper insulation, secure wiring, and safe testing procedures are necessary to prevent electrical hazards and mechanical accidents.
Financial constraints	The project was developed using affordable and locally available components while maintaining acceptable system performance within the budget limits.
Legal restrictions	The project must comply with electrical safety standards and responsible laser usage regulations during testing and implementation.

INTERDISCIPLINARY STUDY FORM

The Interdisciplinary Study form consists of two parts. The first one is the interdisciplinary work organized by the Department Head and in which participation is compulsory. The second is the project support work carried out by student groups or with people from different professions outside the department while the Graduation Project is being carried out. For example, having a carpenter build a table or case for the project.

Disiplinler arası çalışma	Açıklama
Workshops or courses organized by the Department Head	During the graduation project period, participation was made in interdisciplinary workshops and technical activities organized by the department. These activities involved communication and collaboration with students from different engineering departments, which improved teamwork, technical discussion, and problem-solving skills. The experience helped in understanding how mechanical, electrical, and software systems can be integrated together in real engineering applications.
Work done with student groups and / or industrial processes for graduation projects	The graduation project required collaboration between team members in hardware assembly, programming, circuit design, and mechanical integration. Different tasks were distributed among the group members to improve project efficiency and teamwork organization. In addition, support was received from local electronic suppliers and workshop technicians during prototype implementation and component installation. These collaborations provided practical experience in project management, manufacturing, and real-world engineering applications.

OTHER APPENDICES

Arduino code:

```
#include <Servo.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <SoftwareSerial.h>

// =====
// Bluetooth HC-05
// =====
SoftwareSerial BT(A2, A3);
// HC-05 TXD -> A2 Arduino RX
// HC-05 RXD -> A3 Arduino TX

char btCommand = 'S';
bool manualMode = false;
int stopPressCount = 0;
const char* manualMotionText = "STOP";

// =====
// LCD
// =====
LiquidCrystal_I2C lcd(0x27, 16, 2);
unsigned long lastLcdUpdate = 0;
const int lcdInterval = 200;

// =====
// Servos
// =====
Servo scanServo;
Servo armBaseServo;
Servo armUpServo;
Servo armExtendServo;

// =====
// Pins
// =====
const int scanPin = 8;
const int basePin = 5;
const int upPin = 6;
```

```
const int extendPin = 7;

const int laserPin = 9;
const int buzzerPin = 12;

const int trigPin = 10;
const int echoPin = 11;

const int IN1 = 2;
const int IN2 = 3;
const int IN3 = 4;
const int IN4 = 13;

const int irLeftPin = A0;
const int irRightPin = A1;
const int IR_OBSTACLE_STATE = LOW;

// =====
// Arm Positions
// =====
const int baseHome = 77;
const int upHome = 20;
const int extendHome = 160;

const int upTarget = 90;
const int extendTarget = 90;

// =====
// Scanner
// =====
int scanAngle = 100;
int scanDirection = 1;

const int scanMin = 40;
const int scanMax = 170;
const int scanStep = 3;
const int scanDelay = 30;

unsigned long lastScanTime = 0;

// =====
// Detection
// =====
const int detectMin = 4;
const int detectMax = 35;
const int requiredConfirmations = 4;
```

```

int detectionCount = 0;
int detectedAngleSaved = 0;
long lastDistance = 999;
long engagedDistance = 999;

// =====
// Arm Mapping
// =====
const int baseMin = 40;
const int baseMax = 170;

int leftOffset = -25;
int middleOffset = -25;
int rightOffset = -40;

// =====
// Smooth Arm
// =====
float baseCurrent = baseHome;
float upCurrent = upHome;
float extendCurrent = extendHome;

float baseTargetNow = baseHome;
float upTargetNow = upHome;
float extendTargetNow = extendHome;

const float armStep = 0.8;
const int armInterval = 5;
unsigned long lastArmMoveTime = 0;

// =====
// Obstacle Avoidance
// =====
unsigned long avoidStartTime = 0;
int avoidStep = 0;
int avoidTurnDirection = 1;

const unsigned long stopBeforeBackTime = 200;
const unsigned long backTime = 600;
const unsigned long turnTime = 650;
const unsigned long stopAfterTurnTime = 200;

// =====
// System States
// =====
enum SystemState {
    SCANNING,

```

```

    AIMING,
    LOCKED,
    RETURNING_HOME,
    AVOID_OBSTACLE
};

SystemState state = SCANNING;

// =====
// Setup
// =====
void setup() {
    Serial.begin(9600);
    BT.begin(9600);

    scanServo.attach(scanPin);
    armBaseServo.attach(basePin);
    armUpServo.attach(upPin);
    armExtendServo.attach(extendPin);

    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);

    pinMode(laserPin, OUTPUT);
    pinMode(buzzerPin, OUTPUT);

    pinMode(irLeftPin, INPUT);
    pinMode(irRightPin, INPUT);

    pinMode(IN1, OUTPUT);
    pinMode(IN2, OUTPUT);
    pinMode(IN3, OUTPUT);
    pinMode(IN4, OUTPUT);

    digitalWrite(laserPin, LOW);
    digitalWrite(buzzerPin, LOW);
    stopCar();

    scanServo.write(scanAngle);
    armBaseServo.write(baseHome);
    armUpServo.write(upHome);
    armExtendServo.write(extendHome);

    lcd.init();
    lcd.backlight();

    lcd.clear();
    lcd.setCursor(0, 0);

```

```

    lcd.print("EYE OF HORUS");
    lcd.setCursor(0, 1);
    lcd.print("SYSTEM READY");

    startupBeep();
    delay(1500);
}

// =====
// Main Loop
// =====
void loop() {
    updateArmSmooth();
    checkBluetooth();

    if (manualMode) {
        manualControl();
        manualTargetingSystem();
        return;
    }

    if (state == SCANNING) scanMode();
    else if (state == AIMING) aimingMode();
    else if (state == LOCKED) lockedMode();
    else if (state == RETURNING_HOME) returningHomeMode();
    else if (state == AVOID_OBSTACLE) avoidObstacleMode();
}

// =====
// Bluetooth
// =====
void checkBluetooth() {
    while (BT.available()) {
        char command = BT.read();

        if (command == '\n' || command == '\r') continue;

        command = toupper(command);

        Serial.print("BT: ");
        Serial.println(command);

        if (command == 'F' || command == 'B' || command == 'L' || command == 'R') {
            btCommand = command;
            manualMode = true;
            stopPressCount = 0;

            if (state == AVOID_OBSTACLE) state = SCANNING;

```

```

digitalWrite(laserPin, LOW);
buzzerOff();

if (state == SCANNING) {
    baseTargetNow = baseHome;
    upTargetNow = upHome;
    extendTargetNow = extendHome;
}

updateLCD("MANUAL CONTROL", "SURVEILLANCE ON");
}

else if (command == 'S') {
    if (!manualMode) {
        manualMode = true;
        stopPressCount = 1;
    } else {
        stopPressCount++;
    }

    btCommand = 'S';
    stopCar();
    manualMotionText = "STOP";

    if (stopPressCount >= 2) {
        stopPressCount = 0;
        returnToAutoMode();
    } else {
        updateLCD("MANUAL STOP", "SCAN ACTIVE");
    }
}

else if (command == 'A') {
    returnToAutoMode();
}
}

// =====
// Manual Hybrid Mode
// =====
void manualControl() {
    if (state == AIMING || state == LOCKED || state == RETURNING_HOME) {
        stopCar();
        return;
    }
}

```

```
if (isObstacleDetected() && btCommand == 'F') {  
    stopCar();  
    fastBeep();  
    updateLCD("MANUAL SAFETY", "OBSTACLE STOP");  
    return;  
}
```

```
buzzerOff();
```

```
if (btCommand == 'F') {  
    manualMotionText = "FORWARD";  
    moveCarForward();  
}  
else if (btCommand == 'B') {  
    manualMotionText = "BACKWARD";  
    moveCarBackward();  
}  
else if (btCommand == 'L') {  
    manualMotionText = "LEFT";  
    turnCarLeft();  
}  
else if (btCommand == 'R') {  
    manualMotionText = "RIGHT";  
    turnCarRight();  
}  
else if (btCommand == 'S') {  
    manualMotionText = "STOP";  
    stopCar();  
}  
}
```

```
void manualTargetingSystem() {  
    if (state == SCANNING) {  
        if (millis() - lastScanTime >= scanDelay) {  
            lastScanTime = millis();  
  
            scanServo.write(scanAngle);  
  
            long distance = readDistance();  
            lastDistance = distance;  
  
            updateManualScanLCD(distance);  
  
            if (distance >= detectMin && distance <= detectMax) {  
                detectionCount++;  
                detectedAngleSaved = scanAngle;  
  
                if (detectionCount >= requiredConfirmations) {
```

```

        stopCar();

        updateLCD("TARGET ACQUIRED", "MANUAL PAUSED");

        int accurateAngle = findBestTargetAngle(detectedAngleSaved);
        startAiming(accurateAngle);

        detectionCount = 0;
        return;
    }
}
else {
    detectionCount = 0;
}

    updateScanAngle();
}
}
else if (state == AIMING) {
    aimingMode();
}
else if (state == LOCKED) {
    lockedMode();
}
else if (state == RETURNING_HOME) {
    returningHomeMode();
}
}
}

void returnToAutoMode() {
    manualMode = false;
    btCommand = 'S';
    stopPressCount = 0;
    manualMotionText = "STOP";

    stopCar();

    detectionCount = 0;
    engagedDistance = 999;

    digitalWrite(laserPin, LOW);
    buzzerOff();

    baseTargetNow = baseHome;
    upTargetNow = upHome;
    extendTargetNow = extendHome;

    state = SCANNING;

```

```
    updateLCD("AUTO MODE", "MISSION RESUMED");  
}
```

```
// =====
```

```
// LCD
```

```
// =====
```

```
void updateLCD(const char* line1, const char* line2) {  
    if (millis() - lastLcdUpdate < lcdInterval) return;  
    lastLcdUpdate = millis();
```

```
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print(line1);  
    lcd.setCursor(0, 1);  
    lcd.print(line2);  
}
```

```
void updateScanningLCD(long distance) {  
    if (millis() - lastLcdUpdate < lcdInterval) return;  
    lastLcdUpdate = millis();
```

```
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print("AUTO: SCANNING");
```

```
    lcd.setCursor(0, 1);  
    lcd.print("D:");  
    if (distance == 999) lcd.print("---");  
    else lcd.print(distance);
```

```
    lcd.print("cm A:");  
    lcd.print(scanAngle);  
}
```

```
void updateManualScanLCD(long distance) {  
    if (millis() - lastLcdUpdate < lcdInterval) return;  
    lastLcdUpdate = millis();
```

```
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print("MANUAL:");  
    lcd.print(manualMotionText);
```

```
    lcd.setCursor(0, 1);  
    lcd.print("SCAN D:");  
    if (distance == 999) lcd.print("---");  
    else lcd.print(distance);
```

```

    lcd.print("cm");
}

void updateEngagedLCD(long distance) {
    if (millis() - lastLcdUpdate < lcdInterval) return;
    lastLcdUpdate = millis();

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("TARGET ENGAGED");

    lcd.setCursor(0, 1);
    lcd.print("LOCK D:");
    if (distance == 999) lcd.print("---");
    else lcd.print(distance);

    lcd.print("cm");
}

// =====
// Auto Scanning
// =====
void scanMode() {
    buzzerOff();

    if (isObstacleDetected()) {
        startAvoidObstacle();
        return;
    }

    moveCarForward();

    if (millis() - lastScanTime >= scanDelay) {
        lastScanTime = millis();

        scanServo.write(scanAngle);

        long distance = readDistance();
        lastDistance = distance;

        updateScanningLCD(distance);

        if (distance >= detectMin && distance <= detectMax) {
            detectionCount++;
            detectedAngleSaved = scanAngle;

            if (detectionCount >= requiredConfirmations) {
                stopCar();
            }
        }
    }
}

```

```

        updateLCD("TARGET ACQUIRED", "VERIFYING LOCK");

        int accurateAngle = findBestTargetAngle(detectedAngleSaved);

        startAiming(accurateAngle);

        detectionCount = 0;
        return;
    }
}
else {
    detectionCount = 0;
}

    updateScanAngle();
}
}

// =====
// Obstacle Avoidance
// =====
void startAvoidObstacle() {
    stopCar();
    digitalWrite(laserPin, LOW);
    buzzerOff();

    bool leftBlocked = leftObstacleDetected();
    bool rightBlocked = rightObstacleDetected();

    if (leftBlocked && !rightBlocked) avoidTurnDirection = 1;
    else if (rightBlocked && !leftBlocked) avoidTurnDirection = -1;
    else avoidTurnDirection = 1;

    avoidStep = 0;
    avoidStartTime = millis();

    updateLCD("SAFETY ALERT", "OBSTACLE FOUND");

    state = AVOID_OBSTACLE;
}

void avoidObstacleMode() {
    unsigned long now = millis();

    if (avoidStep == 0) {
        stopCar();
        updateLCD("AUTO RECOVERY", "STOPPING");
    }
}

```

```

fastBeep();

if (now - avoidStartTime >= stopBeforeBackTime) {
    avoidStep = 1;
    avoidStartTime = now;
}
}
else if (avoidStep == 1) {
    moveCarBackward();
    updateLCD("AUTO RECOVERY", "REVERSING");
    fastBeep();

    if (now - avoidStartTime >= backTime) {
        avoidStep = 2;
        avoidStartTime = now;
    }
}
else if (avoidStep == 2) {
    if (avoidTurnDirection == 1) {
        turnCarRight();
        updateLCD("AUTO RECOVERY", "TURN RIGHT");
    } else {
        turnCarLeft();
        updateLCD("AUTO RECOVERY", "TURN LEFT");
    }
}

slowBeep();

if (now - avoidStartTime >= turnTime) {
    avoidStep = 3;
    avoidStartTime = now;
}
}
else if (avoidStep == 3) {
    stopCar();
    buzzerOff();
    updateLCD("PATH CLEARED", "SCAN RESUMING");

    if (now - avoidStartTime >= stopAfterTurnTime) {
        detectionCount = 0;
        state = SCANNING;
    }
}
}
}

```

```

// =====
// Target Correction
// =====
int findBestTargetAngle(int centerAngle) {
    int bestAngle = centerAngle;
    long bestDistance = 999;

    for (int a = centerAngle - 6; a <= centerAngle + 6; a += 3) {
        int testAngle = constrain(a, scanMin, scanMax);

        scanServo.write(testAngle);
        delay(35);

        long d = readDistance();

        if (d >= detectMin && d <= detectMax && d < bestDistance) {
            bestDistance = d;
            bestAngle = testAngle;
        }
    }

    scanServo.write(bestAngle);
    delay(40);

    engagedDistance = bestDistance;

    return bestAngle;
}

// =====
// Aiming
// =====
void startAiming(int detectedAngle) {
    stopCar();

    scanServo.write(detectedAngle);

    int baseTarget = map(detectedAngle, scanMin, scanMax, baseMin, baseMax);

    if (detectedAngle < 95) baseTarget += leftOffset;
    else if (detectedAngle > 105) baseTarget += rightOffset;
    else baseTarget += middleOffset;

    baseTarget = constrain(baseTarget, baseMin, baseMax);

    baseTargetNow = baseTarget;

```

```

upTargetNow = upTarget;
extendTargetNow = extendTarget;
digitalWrite(laserPin, LOW);

updateLCD("TARGETING MODE", "ARM ALIGNING");

state = AIMING;
}

void aimingMode() {
  stopCar();
  slowBeep();

  updateLCD("AIMING SYSTEM", "LOCKING TARGET");

  if (armAtTarget()) {
    digitalWrite(laserPin, HIGH);
    updateEngagedLCD(engagedDistance);
    state = LOCKED;
  }
}

void lockedMode() {
  stopCar();
  fastBeep();

  static unsigned long lastCheckTime = 0;

  if (millis() - lastCheckTime >= 100) {
    lastCheckTime = millis();

    long distance = readDistance();
    lastDistance = distance;

    updateEngagedLCD(distance);

    if (!(distance >= detectMin && distance <= detectMax)) {
      digitalWrite(laserPin, LOW);
      buzzerOff();

      updateLCD("TARGET LOST", "RETURNING HOME");

      baseTargetNow = baseHome;
      upTargetNow = upHome;
      extendTargetNow = extendHome;

      state = RETURNING_HOME;
    }
  }
}

```

```
}  
}
```

```
void returningHomeMode() {  
  stopCar();  
  buzzerOff();
```

```
  updateLCD("SYSTEM RESET", "ARM TO HOME");
```

```
  if (millis() - lastScanTime >= scanDelay) {  
    lastScanTime = millis();  
    scanServo.write(scanAngle);  
    updateScanAngle();  
  }
```

```
  if (armAtTarget()) {  
    detectionCount = 0;  
    engagedDistance = 999;  
    state = SCANNING;  
  }  
}
```

```
// =====  
// Smooth Arm  
// =====
```

```
void updateArmSmooth() {  
  if (millis() - lastArmMoveTime < armInterval) return;
```

```
  lastArmMoveTime = millis();
```

```
  baseCurrent = moveSmoothValue(baseCurrent, baseTargetNow);  
  upCurrent = moveSmoothValue(upCurrent, upTargetNow);  
  extendCurrent = moveSmoothValue(extendCurrent, extendTargetNow);
```

```
  armBaseServo.write(round(baseCurrent));  
  armUpServo.write(round(upCurrent));  
  armExtendServo.write(round(extendCurrent));  
}
```

```
float moveSmoothValue(float current, float target) {  
  if (abs(current - target) <= armStep) return target;  
  
  if (current < target) return current + armStep;  
  else return current - armStep;  
}
```

```
bool armAtTarget() {  
  return abs(baseCurrent - baseTargetNow) < 1 &&
```

```

        abs(upCurrent - upTargetNow) < 1 &&
        abs(extendCurrent - extendTargetNow) < 1;
    }

// =====
// Scanner
// =====
void updateScanAngle() {
    scanAngle += scanDirection * scanStep;

    if (scanAngle >= scanMax) {
        scanAngle = scanMax;
        scanDirection = -1;
    }

    if (scanAngle <= scanMin) {
        scanAngle = scanMin;
        scanDirection = 1;
    }
}

// =====
// Ultrasonic
// =====
long readDistance() {
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);

    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);

    digitalWrite(trigPin, LOW);

    long duration = pulseIn(echoPin, HIGH, 30000);

    if (duration == 0) return 999;

    long distance = duration * 0.034 / 2;

    if (distance < 3 || distance > 300) return 999;

    return distance;
}

```

```
// =====  
// IR Sensors  
// =====  
bool leftObstacleDetected() {  
    return digitalRead(irLeftPin) == IR_OBSTACLE_STATE;  
}  
  
bool rightObstacleDetected() {  
    return digitalRead(irRightPin) == IR_OBSTACLE_STATE;  
}  
  
bool isObstacleDetected() {  
    return leftObstacleDetected() || rightObstacleDetected();  
}  
  
// =====  
// Car Movement  
// =====  
void moveCarForward() {  
    digitalWrite(IN1, LOW);  
    digitalWrite(IN2, HIGH);  
  
    digitalWrite(IN3, LOW);  
    digitalWrite(IN4, HIGH);  
}  
  
void moveCarBackward() {  
    digitalWrite(IN1, HIGH);  
    digitalWrite(IN2, LOW);  
  
    digitalWrite(IN3, HIGH);  
    digitalWrite(IN4, LOW);  
}  
  
void turnCarLeft() {  
    digitalWrite(IN1, HIGH);  
    digitalWrite(IN2, LOW);  
  
    digitalWrite(IN3, LOW);  
    digitalWrite(IN4, HIGH);  
}  
  
void turnCarRight() {  
    digitalWrite(IN1, LOW);  
    digitalWrite(IN2, HIGH);
```

```
    digitalWrite(IN3, HIGH);  
    digitalWrite(IN4, LOW);  
}
```

```
void stopCar() {  
    digitalWrite(IN1, LOW);  
    digitalWrite(IN2, LOW);
```

```
    digitalWrite(IN3, LOW);  
    digitalWrite(IN4, LOW);  
}
```

```
// =====  
// Buzzer  
// =====
```

```
void startupBeep() {  
    digitalWrite(buzzerPin, HIGH);  
    delay(150);  
    digitalWrite(buzzerPin, LOW);  
}
```

```
void buzzerOff() {  
    digitalWrite(buzzerPin, LOW);  
}
```

```
void slowBeep() {  
    static unsigned long lastBeepTime = 0;  
    static bool buzzerState = false;  
  
    if (millis() - lastBeepTime >= 400) {  
        lastBeepTime = millis();  
        buzzerState = !buzzerState;  
        digitalWrite(buzzerPin, buzzerState);  
    }  
}
```

```
void fastBeep() {  
    static unsigned long lastBeepTime = 0;  
    static bool buzzerState = false;  
  
    if (millis() - lastBeepTime >= 120) {  
        lastBeepTime = millis();  
        buzzerState = !buzzerState;  
        digitalWrite(buzzerPin, buzzerState);  
    }  
}
```

SHORT BIOGRAPHIES

Amr Ayman Mohamed :

Amr Ayman Mohamed is a senior Electrical and Electronics Engineering student at OSTİM Technical University, interested in **embedded systems, drones, radar, and solar energy**. He has experience with **Arduino, Raspberry Pi, CCS C, MATLAB/Simulink, and FPGA tools** and has worked on VTOL drone and radar projects. He aims to build a career in **embedded systems and renewable energy**.

Ebru Karcı :

Ebru Karcı is a student in the Electrical and Electronics Engineering Department at OSTİM Technical University. During her studies, she has gained foundational experience in electronics, embedded systems, circuit design, and programming, and has participated in various laboratory activities and course projects. Her future goal is to improve her technical skills, gain professional experience, and build a successful career in her field.

Abdalla Hossni

Abdalla Hossni is passionate about sports, especially bodybuilding, which supports his physical fitness, strength, and discipline. He is also interested in engineering and technical design, with experience in AutoCAD, SolidWorks, and FreeCAD for creating accurate drawings and 3D models. Abdalla is continuously developing his design skills to support his future academic and professional career.

Ahmed Soliman

Ahmed Soliman is an Electrical and Electronics Engineering student at OSTİM Technical University, interested in **electronics design, embedded systems, and automation**. He has experience with **MATLAB, Arduino, Proteus, and C programming**, and has completed small sensor-related projects. His goal is to improve his programming and circuit design skills and work in the **automation industry**.

Ahmed Mustafa

Ahmed Mustafa is interested in sports, especially football and bodybuilding, which help him build discipline, fitness, and teamwork. He is also passionate about **electronics and embedded systems**, focusing on microcontrollers and hardware-based solutions. He aims to improve his skills in **embedded programming and circuit analysis** and pursue a career in intelligent system design.

GRADUATION PROJECT I - FILE SUBMISSION CHECKLIST

STUDENT INFORMATION

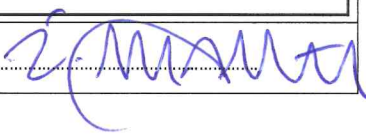
DATE: / /

Number	Name and Last name	Signature
210202885	AHMED MOHAMED IBRAHIM MOSTAFA	
210202967	AHMED MAHMOUD SOLIMAN SOLIMAN	
210202996	AMR AYMAN MOHAMED ELMOATASEMBELLAH	
210202943	ABDALLA ELSAYED HOSSNI MOHAMED AHMED	

This form will be filled out and signed according to the number of students in the project. Students who sign and submit this file accept the accuracy of the information they provide below.

220202019 EBRU KARCI

PROJECT INFORMATION

Project Title	:	The Eye Of Horus
Project Supervisor	:	Prof.Dr. İsmail Hakkı Altaş Signature 

FOR THE PROJECT FILE TO BE SUBMITTED

All of the following answers must be YES.

- Is the number of verbatim quotations from any source in your dossier LESS than 20% and is the number of paragraphs of verbatim quotations zero? YES ☒ NO ☐
- Does your dossier include a preface, abstract, main sections (Introduction, Theoretical Background, Design, Simulation, Experimental Study, Results, Evaluations) and a list of references? YES ☒ NO ☐
- Is there an Introduction, Literature Review, Originality (novelty), Methodology, Impact, List of Standards and Work Schedule in the Introduction section? Is there leadership sharing and B plans in the work packages? YES ☒ NO ☐
- Has a financial analysis commentary and a legal assessment that may arise from the project been added at the end of the design section? YES ☒ NO ☐
- Are all equations numbered, all figures captioned and all charts captioned? YES ☒ NO ☐
- Are all equations, figures, charts and references cited in the text? YES ☒ NO ☐
- Are all the figures in the Design, Simulation and results section yours? YES ☒ NO ☐
- Are all references listed in accordance with the Graduation Project Writing Guidelines and at least 5 scientific articles, theses and dissertations are cited? YES ☒ NO ☐
- Have you included the IEEE Code of Ethics in the Appendix? YES ☒ NO ☐
- Have you conducted interdisciplinary work and explained this in the Appendix? YES ☒ NO ☐

**In order for the project file to be delivered,
all of the above answers must be YES.**

Student Number, Name and Surname : 220202019 - Ebru Karci
: 2. 210202825 - Ahmed Mostafa
: 3. 210202967 - Ahmed Soliman
: 4. 210202996 - Amr Ayman
: 5. 210202943 - Abdalla Hossni
Project title : The Eye Of Horus

Supervisor Title, Name and Surname : Prof. Dr. İsmail Haluk Altas

STUDENT WORK TRACKING SHEET

Education Year 2025 / 2026

Date visited	Status of the Graduation Project	Advisor's signature
Week 1		
Week 2		
Week 3		
Week 4	Konu Belirlenmesi	
Week 5	Prose Araştırması	
Week 6	Kaynak Taraması	
Week 7	Tobitak raporu hazırlığı	
Week 8	Midterm Week	
Week 10	Tobitak Başvurusu	
Week 11		
Week 12	Mezuniyet Prosesi Raporu Hazırlığı	
Week 13		
Week 14		
Week 15		
Week 16		

Öğrencilerin veya çalışma gruplarının haftalık toplantıları sırasında bu formu proje yöneticisine imzalatmaları ve dönem sonunda proje raporuyla birlikte teslim etmek üzere saklamaları gerekmektedir.



GRADUATION PROJECT II - FILE SUBMISSION CHECKLIST

STUDENT INFORMATION

DATE: 8 / 5 / 2026

Number	Name and Last name	Signature
210202996	AMR AYMAN MOHAMED	
210202943	ABDALLA ELSAYED HOSSNI	
210202967	AHMED MAHMOUD SOLIMAN	
210202885	AHMED MOHAMED IBRAHIM	

This form will be filled out and signed according to the number of students in the project. Students who sign and submit this file accept the accuracy of the information they provide below.

220202019

EBRU KARCI

PROJECT INFORMATION

Project Title	:	EYE OF HORUS
Project Supervisor	:	Prof.Dr. Ismail Hakki Altaş
	Signature	

FOR THE PROJECT FILE TO BE SUBMITTED

All of the following answers must be YES.

1. Is the number of verbatim quotations from any source in your dossier LESS than 20% and is the number of paragraphs of verbatim quotations zero? YES ☒ NO ☐
2. Does your dossier include a preface, abstract, main sections (Introduction, Theoretical Background, Design, Simulation, Experimental Study, Results, Evaluations) and a list of references? YES ☒ NO ☐
3. Is there an Introduction, Literature Review, Originality (novelty), Methodology, Impact, List of Standards and Work Schedule in the Introduction section? Is there leadership sharing and B plans in the work packages? YES ☒ NO ☐
4. Has a financial analysis commentary and a legal assessment that may arise from the project been added at the end of the design section? YES ☒ NO ☐
5. Are all equations numbered, all figures captioned and all charts captioned? YES ☒ NO ☐
6. Are all equations, figures, charts and references cited in the text? YES ☒ NO ☐
7. Are all the figures in the Design, Simulation and results section yours? YES ☒ NO ☐
8. Are all references listed in accordance with the Graduation Project Writing Guidelines and at least 5 scientific articles, theses and dissertations are cited? YES ☒ NO ☐
9. Have you included the IEEE Code of Ethics in the Appendix? YES ☒ NO ☐
10. Have you conducted interdisciplinary work and explained this in the Appendix? YES ☒ NO ☐

In order for the project file to be delivered,
all of the above answers must be YES.



2209-A Üniversite Öğrencileri Araştırma Projeleri Destekleme Programı Başvuru Formu

PROGRAM KODU	BAŞVURU DÖNEMİ	SON BAŞVURU TARİHİ	BAŞVURU NUMARASI
2209-A	2025 / 1	12.11.2025 17:30:00	1919B012557172

Kişisel Bilgiler			
TC Kimlik No/Pasaport	99546810686		
Ad Soyad	AMR AYMAN MOHAMED ELMOATASEMBELLAH		
Cinsiyet	ERKEK		
Doğum Tarihi	01/01/2003		
Uyruk	MISIRLI		
İletişim Bilgileri			
Ülke	TÜRKİYE	Şehir	ANKARA
İlçe	KECIOREN	Posta Kodu	06820
Telefon	(055) 245 39 38	Eposta	amorayman9@gmail.com
Adres	Kuzey Ankara Yeşiltepe mahallesi		
Danışmana Ait Bilgiler			
DANIŞMANA AİT BİLGİLER			
TC Kimlik No	10178227390		
Danışman Adı	İSMAİL HAKKI ALTAŞ		
Akademik Ünvanı	PROFESÖR	Görev Ünvanı	PROFESÖR
Çalıştığı Kurum	OSTİM TEKNİK ÜNİVERSİTESİ MÜHENDİSLİK FAKÜLTESİ ELEKTRİK-ELEKTRONİK MÜHENDİSLİĞİ BÖLÜMÜ ELEKTRİK-ELEKTRONİK MÜHENDİSLİĞİ PR. (TAM BURLU)		
Telefon	(533) 559 60 97	Eposta	ihaltas@altas.org
Bütçe			
			Önerilen Bütçe
Ulaşım Giderleri (Uçak ve yataklı tren hariç)			0 TL
Hizmet Alımı Giderleri			0 TL
Makine Techizat Giderleri			5000 TL
Sarf Malzemesi Gideri			3000 TL
Genel Toplam			8000 TL